



Food Insecurity Assessment based on food consumption statistics derived from the Malawi 2004/5 Integrated Household Survey

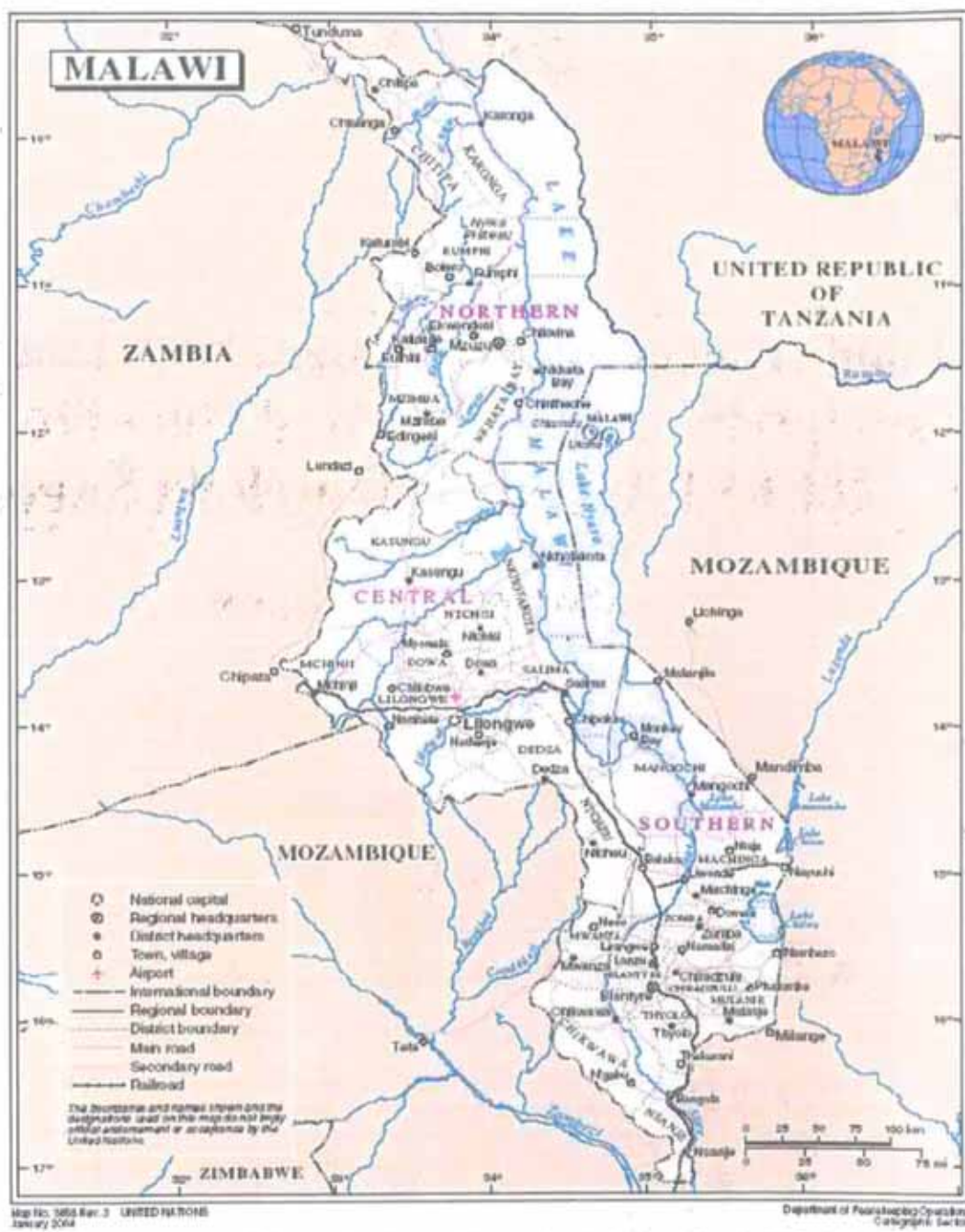
Malawi Country Report



December, 2007

Food and Agriculture Organisation, FAO
National Statistical Office

Map of the Republic of Malawi



Cover page picture: Kilombero Rice farming Karonga district, Malawi (Photo by Willie Kachaka)

Food Insecurity Assessment based on food consumption statistics derived from Malawi IHS2, 2004/5

TABLE OF CONTENTS

GLOSSARY	iv
Executive Summary	vii
I. INTRODUCTION	1
II. BACKGROUND OF THE SURVEY	3
(a) Magnitude of food deprivation	7
(b) Depth of hunger	9
(c) Critical Food poverty	12
(d) Food consumption and expenditures	13
(e) Dietary Diversity	26
(f) Food and income inequality	31
(g) Progress towards the WFS and MDG	35
IV. Conclusion	37
V: Recommendations:	38
V. REFERENCES	38
ANNEXES:	40
<i>Annex Table1. Selective Statistics of food consumption</i>	40
<i>Annex Table 2. Food deprivation at national and sub national levels</i>	41
<i>Annex Table 3. Share of food consumption in monetary value by food sources</i>	42
<i>Annex Table 4. Share of dietary energy consumption by food source</i>	43
<i>Annex Table 5. FAO indicators on food deprivation and food poverty</i>	44
<i>Annex Table 6. Contribution of in grams of nutrient consumed in 1000 calories</i>	45
<i>Annex Table 7. Dispersion ratios of income, food and total consumption</i>	46
<i>Annex Table 8: Inequality measures of food consumption and income, Gini coefficients</i>	47
<i>Annex Table 9. Inequality measures of food consumption and income – Coefficient of variation</i>	49
<i>Annex Table 10: Food demand elasticity with respect to income</i>	51
<i>Annex Table 11: Diet composition - Nutrient consumption</i>	52
<i>Annex Table 12: Total energy and macro nutrient by food commodity group</i>	53
<i>Annex Table 13: Share of nutrients by food commodity group</i>	54

GLOSSARY

BALANCED DIET

The diet is balanced when is judged to be consistent with the maintenance of health in a population. The balance can be examined in terms of the contributions of the various energy-supplying macronutrients and other nutrients. A macronutrient-based balance food consumption pattern should contribute to total energy from proteins, fats and carbohydrates within recommended ranges as follows: proteins from 10 to 15 percent, fats from 15 to 30 percent and carbohydrates from 55 to 75 percent as from a technical report of a 2002 joint WHO/FAO Expert Consultation (WHO 2003).

CRITICAL FOOD POVERTY

The prevalence of critical food poverty (PCFP) refers to the proportion of persons living on less than the cost of the macro-nutrient balanced MDER (for Minimum Dietary Energy Requirement see below and for balanced diet see above) with food prices from households in the lowest income quintile. It can be estimated at national and sub-national levels.

DIETARY ENERGY UNIT COST

The dietary energy unit cost is the monetary value of 1000 kilo-calories of food consumed.

DEPTH OF FOOD DEPRIVATION

It refers to the difference between the average dietary energy consumption of an undernourished population and its average minimum energy requirement (MDER).

DIETARY ENERGY CONSUMPTION

Food consumption expressed in energy terms. At national level, it can be calculated from the FBS (see below); the IHS estimate refers to both private and public food consumption. At sub-national levels is estimated using food consumption data in quantities collected in national household surveys; the IHS estimate refers to private food consumption.

DIETARY ENERGY DEFICIT

Same as Depth of Food deprivation

DIETARY ENERGY REQUIREMENT

It refers to the amount of energy required by an individual to maintain body functions, health and normal physical activity.

The *minimum* dietary energy requirement (MDER) refers to the amount of energy considered adequate to meet the energy needs for normative *minimum* acceptable weight for attained height while performing *light* physical activity in good health.

The *average* dietary energy requirement (ADER) refers to the amount of energy considered adequate to meet the energy needs for normative *average* acceptable weight for attained height while performing *moderate* physical activity in good health.

FOOD BALANCE SHEETS

The food balance sheets (FBS) are derived for each commodity using data on food production and imports and opening-year food stocks after deduction of food export and end-year food stocks and all non-food consumption (animal feed, industrial use, seed, wastage and other non-food use); the IHS estimate refers to both private and public food consumption.

FOOD CONSUMPTION DISTRIBUTION

Food consumption distribution refers to the variation of consumption within a population. It reflects both the disparities due to socioeconomic factors and differences due to biological factors, such as sex, age, body weight and physical activity levels.

FOOD DEPRIVATION

Food deprivation refers to the condition of people whose food consumption is continuously below its requirements. FAO's measure of food deprivation refers to the proportion of the population whose dietary energy consumption is below the MDER.

FOOD INSECURITY

A situation when people lack secure access to sufficient amounts of safe and nutritious food for normal growth and development and an active and healthy life. It may be caused by the unavailability of food, insufficient purchasing power or inappropriate distribution. Food insecurity may be chronic, seasonal or transitory.

FOOD SECURITY

A situation that exists when all people, at all time, have physical, social and economic access to sufficient, safe and nutritious food that meets their dietary needs and food preferences for an active and healthy life.

GINI COEFFICIENT

The Gini coefficient is a measure of inequality and ranges from 0 (perfect equality) to 1 (perfect inequality). In the IHS document it refers to inequality of income.

GINI COEFFICIENT DUE TO INCOME

The Gini coefficient is a measure of inequality in food consumption due to income and ranges from 0 (when income has no effect on food consumption) to 1 (when food consumption depends only on income). In the IHS document it refers to inequality in food expenditure in monetary or in energy terms.

HOUSEHOLD CONSUMPTION EXPENDITURE

Total household consumption expenditure as defined in the United Nations guidelines is the sum of all monetary value or expenditure on goods and services intended for consumption, goods produced and consumed from own production or own-business stocks, including the imputed rent of owner-occupied housing, and goods and services received in kind.

HOUSEHOLD FOOD CONSUMPTION EXPENDITURE

Household consumption expenditure refers to food consumed by household members during a specified period, at home and outside the home, for example, at restaurants, bars, the work place, school, and so on. It includes food from all sources, purchased or from garden or farm. Further deductions should be made to allow for wastage and losses occurring from acquisition to cooking and plate and kitchen wastage.

HOUSEHOLD NON CONSUMPTION EXPENDITURE

It refers to income taxes, other direct taxes, pension and social security contributions, remittances, gifts and similar transfers made by the household in monetary terms or in kind, including food such as given away raw or ready to eat.

HOUSEHOLD EXPENDITURE

Consumption plus non-consumption expenditure made by the household, including food.

HOUSEHOLD INCOME

Income is the sum of all receipts, in money or in kind, which as a rule are received regularly and are of recurring nature, including food.

INCOME ELASTICITY OF FOOD DEMAND

The income elasticity of food demand measures the responsiveness of the food demanded (quantity, monetary or nutrient terms) to a unit change of income.

INCOME INEQUALITY

Income inequality refers to disparities in the distribution of income.

INEQUALITY IN FOOD CONSUMPTION DUE TO INCOME

The inequality refers to the variation of the food consumption level within a population due to disparities in the income distribution.

KILOCALORIE (Kcal)

Kilocalorie is a unit of measurement of dietary energy. In the International System of Units (ISU), the universal unit of dietary energy is the joule (J) but Kcal is still commonly used. One kilocalorie = 4.184 kilojoules (KJ).

MICRONUTRIENTS

The vitamins, minerals and certain other substances required by the body in small amounts. They are measured in milligrams or micrograms.

NUTRITIONAL STATUS

The physiological state of an individual that results from the relationship between nutrient intake and requirement and from the body's ability to digest, absorb and use these nutrients. Lack of food as well as poor health and sanitation and inappropriate care and feeding practices are the major causes of poor nutritional status.

SHARE OF FOOD EXPENDITURE

The proportion of household consumption expenditure allocated to food; it is also known as Engel ratio.

UNDER NOURISHMENT

Same as Food Deprivation.

Executive Summary

This report provides information about Malawi's food insecurity situation at the national and sub national levels, using FAO methodology based on food consumption statistics derived from data collected in the 2004/5 Integrated Household Survey (IHS2).

Although the Malawian economy has improved significantly during the last decade, food insecurity is still quite widespread among its population. In 2004, the estimate of food deprivation, which is the measure of MDG Indicator 5 on prevalence of undernourishment, was 27 percent; it was more in the rural areas 28 percent and 22 percent in the urban population.

Food dietary energy consumption at national level was estimated at an average of 2260 kcal/person/day. The urban population had higher dietary energy consumption (2500 kcal) than the rural population (2230 kcal).

The minimum dietary energy requirement (MDER) based on height, age, sex and body mass index (BMI) was at national level of 1628 kcal/person/day. The MDER in the urban and rural population was 1696 and 1619 kcal/person/day respectively.

The prevalence of critical food poverty, which is a measure of food income deprivation, was 14.4 percent at national level (proportion of population with insufficient income to acquire a balanced food basket yielding the MDER).

The dietary energy consumption of the food insecure population was of 1253 kcal/person/day) and the depth of hunger, which is the shortfall to the MDER (1696 kcal/) was of 375 kcal/person/day.

The share of food monetary value in total consumption was about 50 percent at the national. This was also around the level in all income quintiles and other population groupings. The national average daily per person food expenditure was 45.8 MK and the cost of 1000 kcal dietary energy 16.8 MK.

Malawians acquired 45 percent of their food consumption from purchases, and 43 percent acquired their food consumption from their own production and this pattern was far higher in rural areas (51 percent) probably due to availability of land to the population. In urban areas, purchases were almost the only source of food acquisition (about 88 percent).

At national level 75.5 percent of dietary energy was derived from carbohydrates, 14.9 percent from proteins and 9.5 percent from fats, showing an imbalanced diet. The pattern is up the WHO set standards of min of 55-75 on carbohydrates, 10-15 for proteins and 15-30 for fats. In this case the Malawian diet is within WHO standards on carbohydrates and proteins and on the minimum on the fats requirements.

Nationwide the inequality in access to food due to income measured as the coefficient of variation of dietary energy consumption (CV) due to income at national level was 36.5 percent while Gini coefficient of food dietary energy consumption due to income was 18.9 percent.

I. INTRODUCTION

Food security exists when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food to meet their dietary needs and food preferences for an active and healthy life.

The issue of hunger and food insecurity as cause and effect of poverty has been the concern of international communities to concentrate urgent efforts to alleviate poverty at the country, regional and global levels. Objectives of the World Food Summit and the Millennium Development Goal (MDG) set halving respectively the number and the proportion of people who suffer from hunger by 2015. FAO is the lead agency for monitoring the process in achieving the proposed targets at the global level using country available food data and other complementary data on food requirement and access to food from national surveys collecting food consumption data.

FAO Statistics Division has developed the Food Security Statistics Module (FSSM) software to help National Statistical Organizations (NSO) to optimise the analysis of the existing NHS food consumption data for estimating the prevalence of undernourishment and a suite of food security indicators at national and sub national levels. This initiative has the objective to improve the quality, consistency and availability of key development indicators useful by the national and international communities for evaluating and monitoring progress towards hunger reduction programmes. In addition, it provides a harmonized methodological procedures and recommendations on the compilation of food security statistics (FSS) for comparability purposes at national, regional and global levels. This FSS analysis activity falls under the integrated programme of the EC-FAO food security information for action project aiming to make available the tools, materials and skills for deriving a suite of food security indicators to better identify and locate the food insecure population and their effective use in the fight against hunger.

The FSS training activity was conducted successfully at the IDC at the Food and Agriculture Organisation, headquarters in Rome Italy from 15th July-10th August. The Malawi country team was represented by Mr. Willie Kachaka, Chief Statistician from the National Statistical Office, Mrs. Memory Mtembezeka, Chief Nutrition Officer, from the Department of Nutrition, HIV and AIDS in the Office of the President and Cabinet (OPC), Mr. Chancy Chilimbila, Monitoring and Evaluation officer, from the Ministry of Agriculture and Food security.

1. BACKGROUND

(a) MALAWI AND ITS ECONOMY.

Malawi is one of the countries in the Sub Saharan Africa. It shares boundaries with Tanzania in the Northern part, Zambia in the Western part and Mozambique in the Eastern & Southern part of the country. It has about 119, 140 square kilometres with a current population estimate of 13.8 million people. The population growth rate of Malawi is estimated at 3.3%. The urban rural population distribution shows that 87 percent of the population resides in the rural areas and 13 percent resides in the urban areas.

Agriculture is the mainstay of the country's economy, contributing about 35% of the country's GDP. After a decade of economic downturns reaching a low GDP growth rate of 2.2 percent in 2005, the economy is on its recovery path registering an estimated GDP growth rate of 8.6 in 2006. The trend is expected to improve further with the bumper cereal crop realised in the 2006/07 cropping season, which has eased the pressure of food costs, thereby pushing the inflation to its downward trend. The average annual inflation rate for 2006 was 14.2 percent and this expected to continue its downward trend.

The country's agricultural sector employs about 75 percent of the labour force and has mainly two sectors namely the Estate and Smallholder sectors. Tobacco is the main export earner seconded by Tea. The food security diversification is still narrow, Malawian households rely on a single staple, Maize. Diversification to other foods such as rice, cassava and potatoes is sporadic. Malawi still has some problems with the agro processing and storage of agricultural products which results into exporting mostly unprocessed products.

This report presents the different Food Security Statistics obtained from food consumption analysis of the 2004/05 Malawi Integrated Household Survey (IHS2) and may be used as a resource document for raising awareness of the food insecurity situation for more focused intervention by policy makers and other stakeholders of food security issues. The report provides a brief background the food consumption data as collected from the IHS2 survey. The MDG indicator 5, prevalence of undernourishment together with indicators on critical food poverty and depth of hunger are discussed in the subsequent following sections. The food consumption patterns in terms of dietary energy and monetary are analysed and is followed by an assessment of the diet diversity in terms of the quality and type of food consumption in Malawi. Inequalities estimates such as coefficient of variation (CV), Gini coefficients and dispersion ratios that provide an assessment of access to food are also discussed. Finally, the income demand elasticity of food consumption in terms of expenditure and dietary energy is examined. As the objective of the survey was not meant for a comprehensive food security analysis, the last section gives some limitations of the available food consumption data and the related recommendations for its improvement for more reliable and consistent FSS estimates.

II. BACKGROUND OF THE IHS2 SURVEY

(a) Scope and purpose of the survey

In 1997-98, Malawi conducted the first Integrated Household Survey (IHS-1). The purpose of the study was to provide a complete and integrated data set to better understand target groups of households affected by poverty. In addition, the IHS-1 was to serve a much broader set of applications on policy issues regarding households' behaviour and welfare, distribution of income, employment, health and education.

In 2004, the Government of Malawi decided to conduct the second IHS (IHS-2) in order to compare the situation in 2004 with the situation in 1997-98, and to collect more detailed information in specific areas. This survey was undertaken by the National Statistical Office and it collected information from the nationally representative sample of 11,280 households. The survey was carried out for twelve months from March 2004 through March 2005. The purpose of the survey was to provide information on the various aspects of household welfare in Malawi. The data from this survey has been used as an input to the Food Security Assessment report.

(b) Sample design and selection

The sample design for the IHS-2 was different from the sample design in IHS-1. In both Surveys, the sample was designed to provide district estimates of welfare indicators. Because a Census had been done in 1998 after the IHS-1, it was possible to have an updated sample frame for the sample design used in the IHS-2. The sample for IHS-2 was drawn using a two-stage stratified sampling procedure from a sample frame using the 1998 Population Census Enumeration Areas (EAs).

The sample frame included all the three regions of Malawi: north, centre and south. The IHS-2 Stratified the country into rural and urban strata. The urban strata included the four major urban areas: Lilongwe, Blantyre, Mzuzu, and the Municipality of Zomba. All other areas including Bomas were considered as rural areas. The total sample was 11,280 households (564 EAs x 20 Households). Information on sampling errors for consumption from the IHS1 (October 1997 – September 1998) was used to help determine the minimum sample size in each domain.

These domains were further divided into a number of smaller strata based on the administrative system of the country. Each of the twenty-seven districts was considered as a separate sub-stratum of the main rural stratum (for IHS-2, Likoma District was excluded because of transportation problem to the island, so only twenty-six administrative districts were considered). Thus the total number of strata in the survey was thirty: twenty-six districts and four urban centres.

(c) Survey Questionnaires

The IHS-2 survey used two questionnaires to collect the information: a household Questionnaire; and a community questionnaire, as was the case for the 1997-98 IHS-1. Both the household and community questionnaires were significantly revised for the IHS-2. The IHS-2 household questionnaire maintained comparisons with the earlier IHS-1 Household questionnaire wherever possible. However, the IHS-2 questionnaire was longer and more detailed. In addition, new modules were added.

(d) Survey Implementation

During the survey implementation; 20 households were selected from each sampled PSU/ Enumeration area. The selected household was interviewed once where necessary for all the modules. The interviewer could revisit the household later to conduct anthropometric measurements if there was child aged 5 years and below. The interviewing from one PSU to the other was on a monthly rotation and the sample was equally spread across the 12 months. Household expenditure data was collected on three modules food consumption in seven days recall, non food expenditures 30 days recall, durables 12 months recall period.

(e) Food consumption data

The IHS2 collected a wide range of household consumption expenditures including food and non-food consumption. The collection format mostly adapted the UNSD/COICOP classification of expenditures.

The main source of food consumption data is *module i*. This collected food consumed by the household in the seven days prior to the survey by recall method. The information provided included food sources (purchase, from own production, gifts etc.). Data on food consumed outside from home was collected by classifying some of the foods that are likely to be consumed away from home. For all purchased foods the monetary value of the quantity consumed was collected. Whilst for the food from own source and gifts only quantities were collected.

Unit of Measurement: Whenever a household had consumed food, it reported the quantity of food consumed in a particular unit of measurement. There were 23 different types of units, however in the end all the different items had to be converted to a standard unit that is "*grams*".

(f) Data Limitations

The IHS2 food consumption data had some problems of its own , that are worth noting and improving in the future, such as:

a) **Unit of measurements;**

Conversion of non conversional units proved difficult e.g. heaps and bags of Cassava, potatoes, the weight in grams differ according to variety and region.

In the north the cassava roots are huge and are used to produce cassava flour, in the other regions, the cassava is of smaller and sweeter varieties used for cooking.

Fish; problems of measuring weight of different types of dried fish and fresh fish, or even when the fish type is known; whether the fish was small, medium or large, and what the weight in grams was.

b) **No Quantity or monetary values for some purchased items;**

In most cases where the quantity in standard unit was not reported and the monetary value reported, an effort was made to use unit price values for the particular areas to re-estimate the quantity. There was still very big dispersion of purchased price for food items;

c) **Cooked food eaten away from home** had no quantity measurements and difficult to convert.

d) Reported bulk food purchases that were meant for stock and not for consumption within the seven days recall period.

e) Malawi does not have a **National Food Composition Table**, the food conversion into Kcal energy equivalents used the Africa Food Composition Table, however some food items were not available in the FCT and the USDA was used, but this should be treated with caution as the nutrient values with the USDA may be completely different.

f) Income data was not used it was seen as underestimated, as it was below the reported expenditures, therefore expenditure data was used as a proxy for income.

g) Methodology

The calculation of the statistics was based on the FAO Food Security Statistics Module (FSSM). Attached as Annex 2:

III. RESULTS AND FINDINGS

The food consumption data together with some complementary household and household member's data were analysed using the FSSM software, developed by the Statistic Division of FAO. The FSSM is interactive software having the processing, statistical analysis and integration programmes as separate modules for ease of execution of the different stages preceding the final derivation of the food consumption and security statistics. The food consumption, household and household member's data were extracted from the main IHS2 data files and prepared in three distinct data files according to some predefined formats.

The macro-nutrient conversion factors for the food items consumed during the survey were adapted from the Africa Food Consumption Table (AFCT)¹. In very rare cases where the food items were not identified in the AFCT, corresponding description and conversion were identified from the USDA Food Consumption Table to convert all food quantities into the macro nutrient values, dietary energy, protein, fat and carbohydrate, fibre, ash and alcohol. The FSS¹ derived from the IHS2 are presented below at national and sub national levels with reference to some functional population groupings such as geographical location (urban/rural), the size of the household, gender, economical status and educational level of head of household. The data concerning regions are not presented since they are considered as unreliable due to insufficient sample size. The following major food security indicators are covered:

- *Prevalence of undernourishment*
- *Depth of hunger*
- *Food and critical poverty*
- *Food availability*
- *Inequality of access to food*
- *Diet composition*
- *Income demand elasticity*

¹ The African food consumption table, 1968, and the USDA food consumption table

¹ Sibrian R Ramasawmy S and Mernies J (2007). Measuring hunger at sub national levels from household surveys using the FAO approach: MANUAL. FAO Statistics Division Working Paper Series No. ESS/ESSA/005e. http://www.fao.org/es/ess/faostat/foodsecurity/Papers_en.htm

(a) Magnitude of food deprivation

Food insecure people are those individuals whose food intake falls below their minimum dietary energy requirements.

Millennium Development Goal on hunger targets to halve, between 1990 and 2015, the proportion of people who suffer from hunger and is specified as Indicator 5:

"Proportion of population below minimum level of dietary energy consumption "

The Minimum Dietary Energy Requirement (MDER) is the amount of dietary energy of an average individual that is considered adequate to meet his daily energy needs for light activity and good health given his/her age and sex.

The magnitude of hunger (food deprivation) as measured by the prevalence of under nourishment showed that in the year 2004 to 2005 about one person out of three (27 percent) was undernourished in Malawi. The data on food consumption estimated the MDER for the average Malawian, the estimated minimum dietary energy needed to maintain body-weight and perform a sedentary light physical activity taking into account of the height, age and sex structure of the population. The MDER was 1696 kcal per person per day at national level in 2004/05. The average Dietary Energy Consumption for Malawians was estimated to be 2260 kcal/person/day. About forty one percent of the population had food consumption below the minimum dietary energy requirement (MDER) of the average Malawian as defined by FAO.

Height data was collected for all children aged 5 years and below, in the IHS2, in addition height for the average female and aged 15-49 and their Body Mass Index (BMI) was obtained from the Malawi Demographic and Health Survey conducted during the same year 2004/2005. The height, BMI was used with the age and sex structure of the sampled households' members to estimate the MDER at national and sub national groups.

The level of income is an important determinant of the food deprivation levels, with the level of food deprivation decreasing with an increase in income levels. The level of food deprivation for the lowest income group was quite high 75 percent, whilst in the higher income group 0 percent was undernourished (Figure 1).

Rural areas have a significantly high prevalence of undernourishment of 28 percent compared to a level of 22 percent for the urban areas where only 13 percent of the population resides.

In terms of household size, larger households' in terms of number of members, having six or more persons had a high level of undernourishment of 43 percent. Households with one or two persons had the least level of under nourishment of 3 percent.

The gender disparity in food deprivation level reveals that contrary to the stereotype that female headed households are insecure, the male headed households have a higher level of food deprivation 28 percent compared to female headed households 2 percent. This result could be explained by the cultural dimension in Malawi that females are mostly involved in food preparation and food acquisition.

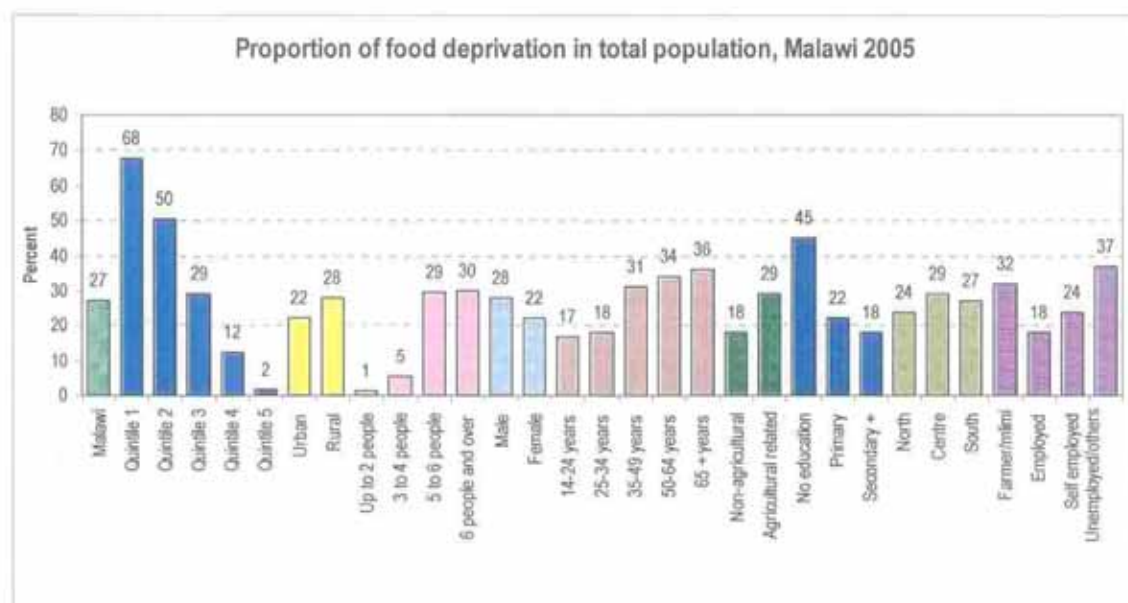
In terms of the household occupation, agricultural related activities was the main occupation of household heads households 80 percent of the households heads were involved in an agricultural related occupation, while only 20 percent were in the non agricultural sector². The variation in terms of food deprivation however, is contrary to the expected distribution. The finding shows that households with a head whose occupation is in agricultural sector had a higher level of food insecurity and hence higher values of prevalence of under nourishment 29 percent, than those occupied in a non agricultural related sector– 18 percent.

In terms of education level, households headed by an individual with secondary education or higher had lower levels of food insecurity, and their level of under nourishment was 18 percent as against 45 percent for the households with a head with no education.

The spatial variation shows that households located in the central region had the highest level of food deprivation levels, 29 percent; the north had the lowest 24 percent, while the south had 27 percent. This is line with the average land holding size variations across the regions. The result for the centre is striking since it's an agricultural area, however the higher levels could be explained by the food diversification which is lower in the central region.

The households heads economic activity also influences the level of food deprivation; households whose heads were unemployed had the highest level of deprivation 37 percent, followed by households involved in farming (farmers) 32 percent, the level of deprivation for the self employed was 24 percent and, the least were households with employed household heads 18 percent.

Figure 1: Prevalence of food deprivation in total in total population

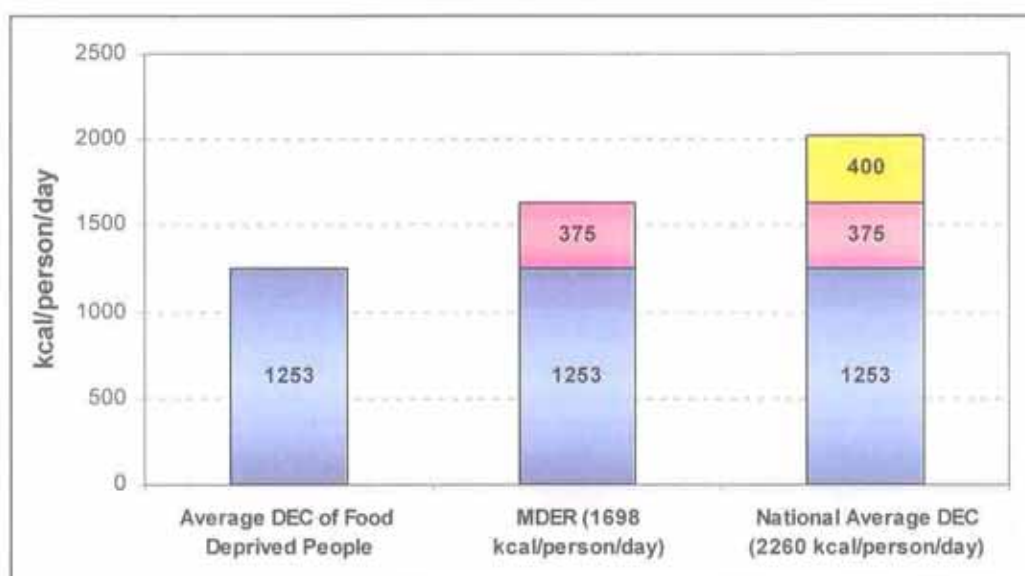


² Households classification occupation whether in an agricultural related activity or a non agricultural related activity, IHS2

(b) Depth of hunger

The 27 percent of undernourished people of Malawi had on average, a daily per person food consumption of only 1253 kcal, which was well below the national MDER benchmark of 1698 kcal/person/day and the national average dietary consumption (DEC) of 2260 kcal/person/day. It then required an additional daily per person of 375 kcal for these people to reach the MDER value. However, to reach the national average dietary energy consumption, an additional 775 kcal was needed (see Figure below).

Figure 2: Depth of Hunger _Malawi 2005



The population of urban areas had a higher average DEC of 2500 kcal, compared to their rural compatriots which was 2250 kcal of rural regions (see Figure below). Similarly, the urban food deprived people had higher average daily food dietary energy consumption (1373 kcal) than those of the rural areas (1284 kcal). However, the urban population that is food deprived had a higher daily food deficit with respect their average DEC (2500) of 436 kcal compared to a value of 396 kcal for the rural population to meet their average DEC (2250kcal). On the other hand the urban food deprived required an additional 323kcal to meet their MDER (1698 kcal), which is lower than the requirement for the rural deprived which required an additional 335 kcal to meet their MDER of (1619 kcal).

Figure 3: Depth and Intensity of hunger, in Urban and Rural, Malawi, 2005

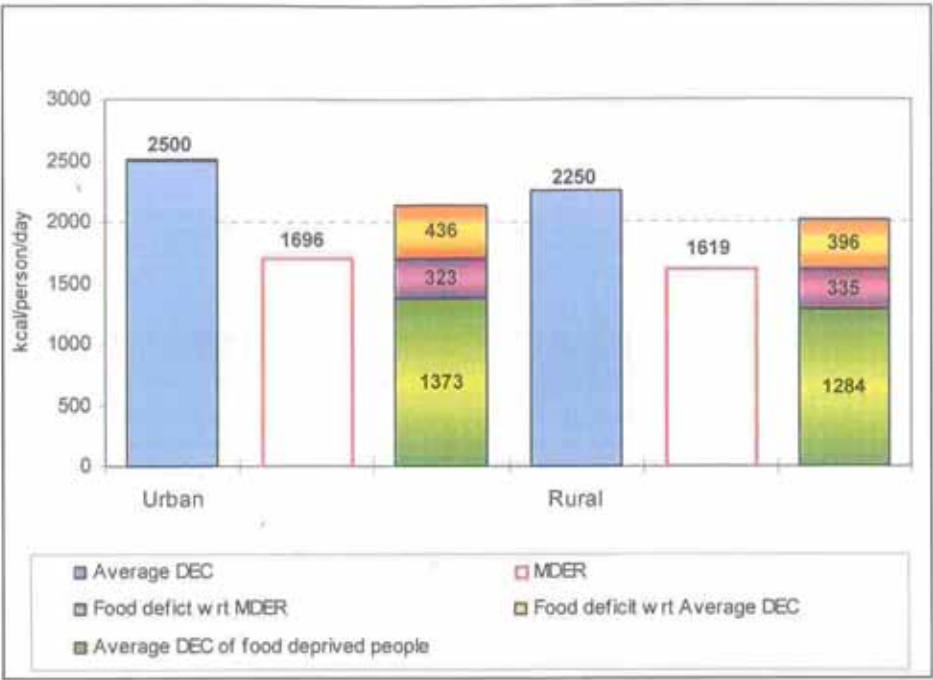


Figure 3 compares the dietary energy consumption and needs of the geographical sub national population groupings to the dietary energy consumption of their food unsecured population within these sub national levels groupings. The minimum dietary energy needs did not vary much among the regions the MDER was lower in the north 1616 kcal, 1628 kcal in the centre and 1630 kcal in the south. However, the average dietary energy consumption showed a large range of variation of 138 kcals with the highest DEC in the north (2349 kcal) and the lowest in the central region 2210 kcal. This region is exceptional as it is the agricultural region of the country but it has the higher prevalence of food deprivation of 29 percent.

The food insecure population DEC had a very lower variation. The centre region however had the lowest DEC of the food deprived of 1288 kcal, compared to 1295 in the north and south regions. See figure below.

Figure 4: Depth of hunger by sub national population (geographical), Malawi 2005

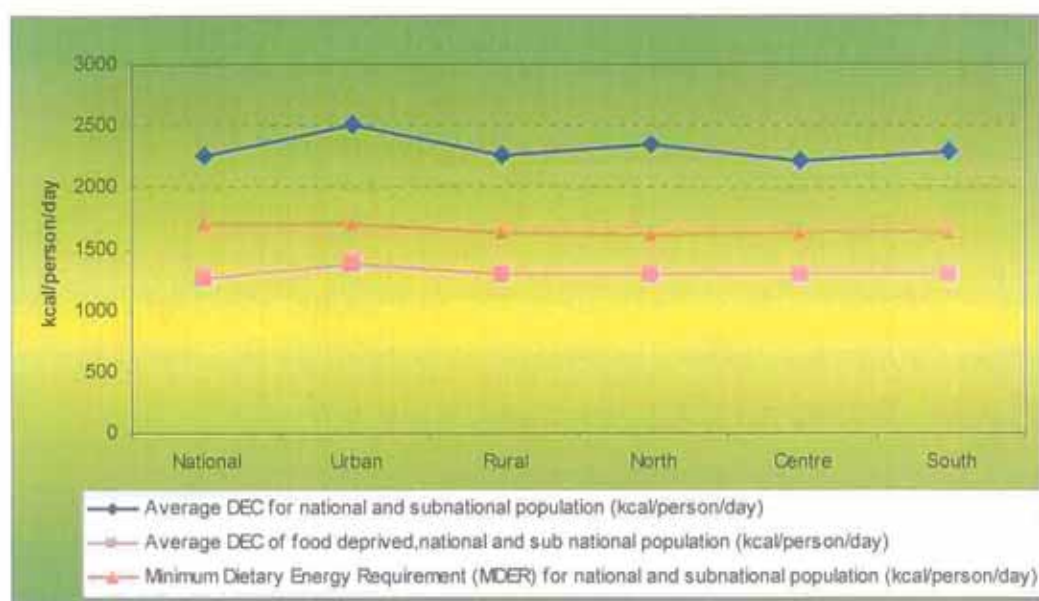
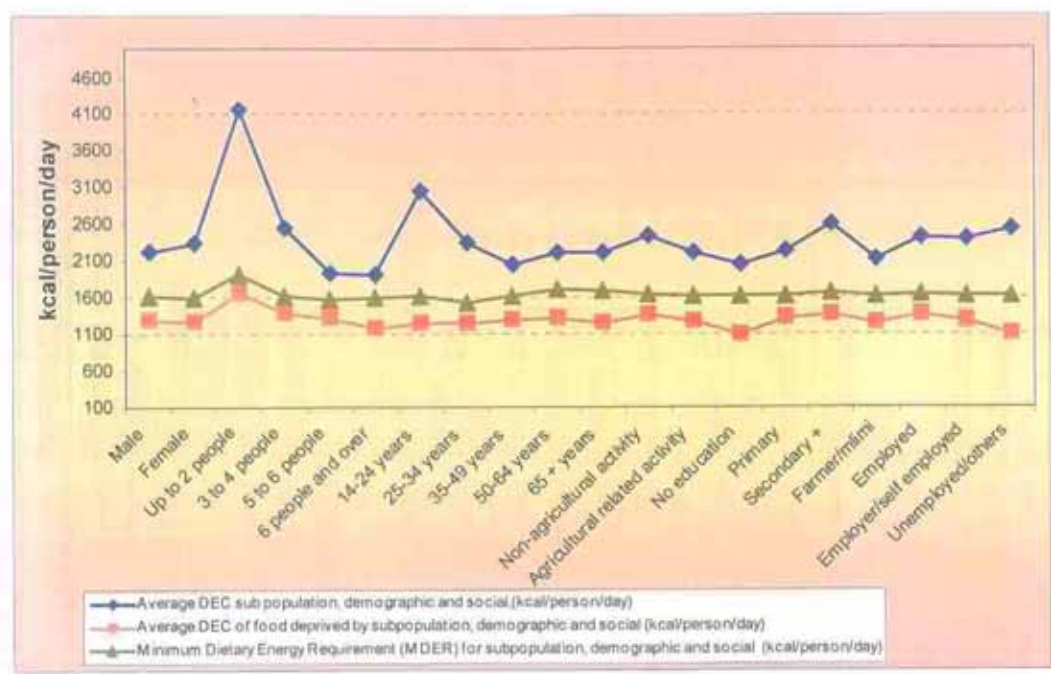


Figure 4 shows the disparities among demographic and socio economic sub national population groupings. The range of variation in the average DEC of the food insecure population was of smaller magnitude than those among the geographical regions. Male headed households were those having a higher DEC for the food deprived households of 1299 kcal compared to female households of 1293 kcal. Small households with up to two persons had the highest DEC, and the food deprived had their DEC 1681 kcal almost close to national MDER of 1628 kcal. Large households having more than four members had average DEC that was quite lower for the food deprived of 1200 kcal. The same is true with the population of those households did not have any employment (unemployed) those households whose heads do not have education. These had an average DEC of those with food deficit of 1094 kcal and 1097 kcal respectively.

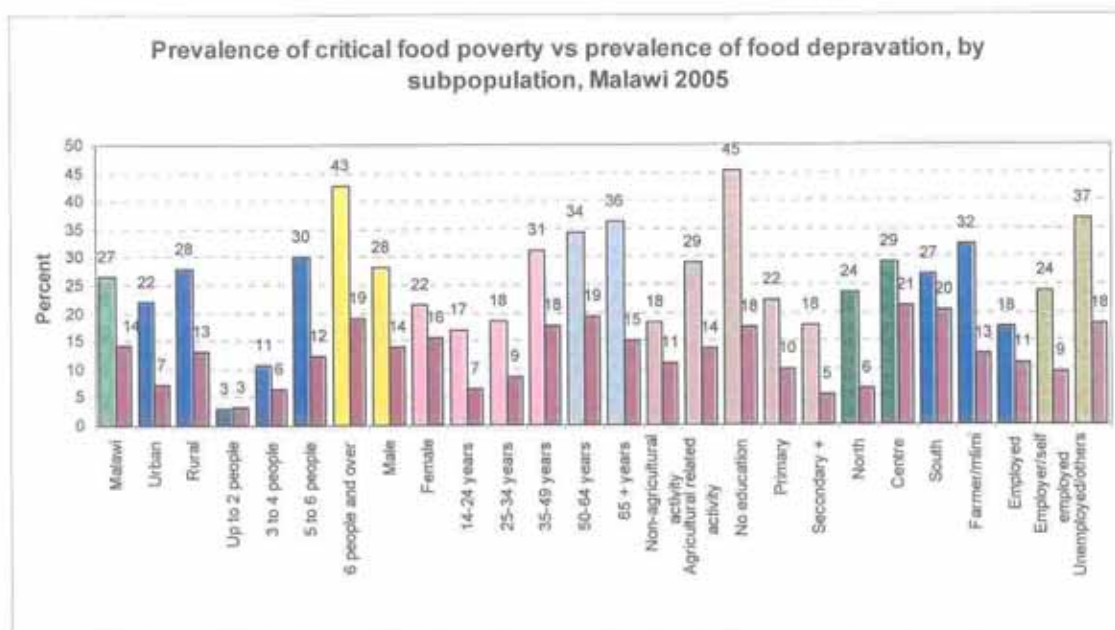
Figure 5: Depth of hunger by subpopulation (demographic and social characteristics), Malawi 2005



(C) Critical Food poverty

The prevalence of Critical Food Poverty (pCFP) is defined as the proportion of the population whose income is lower than the cost of a macronutrient-balanced food basket equivalent to the minimum dietary energy requirement (MDER). The MDER cost is valued using macro-nutrient unit costs from food consumed by households in the first income quintile. About 14.4% of the total population of Malawi were in critical food poverty (Food Insecure). The survey further showed that 13.2% of the rural people were in critical food poverty as compared to 7.2% of their urban counterparts. As the household size was increasing the critical food poverty was getting worse as evidenced by the higher critical food poverty of 19.1% of the households with over six members whilst those households that had two members had the lowest critical food poverty of 3.3%. There was no major variation in food poverty between the gender categories by the survey period, (Figure 2).

Figure 6: Prevalence of Critical Food Poverty



The survey also revealed that there were less food insecure households in the northern part of Malawi if compared to the other regions. There were 6.4% households of the population in the northern region that were in critical food poverty. This was due to the wide variety of food items such as rice, maize, cassava, bananas etc that are being used as food in the northern part of Malawi whilst the central region of Malawi registered the highest number of (21.1%) people that were in critical food poverty this was particularly due to reliance of one food crop (maize). In general terms it can be said that the Northern part of Malawi was Food secure by the time of the survey period. The unemployed households were mostly affected with 18.0% of the population being in critical food poverty. This was due to the fact that the unemployed households mostly relied on food from own production and once it was completed there were food insecure.

(D) Food consumption and expenditures

1. Dietary Energy consumption (DEC)

The calculated Minimum Daily Energy Requirement (MDER) for an average Malawian was calculated at 1628 kcal/person/day.

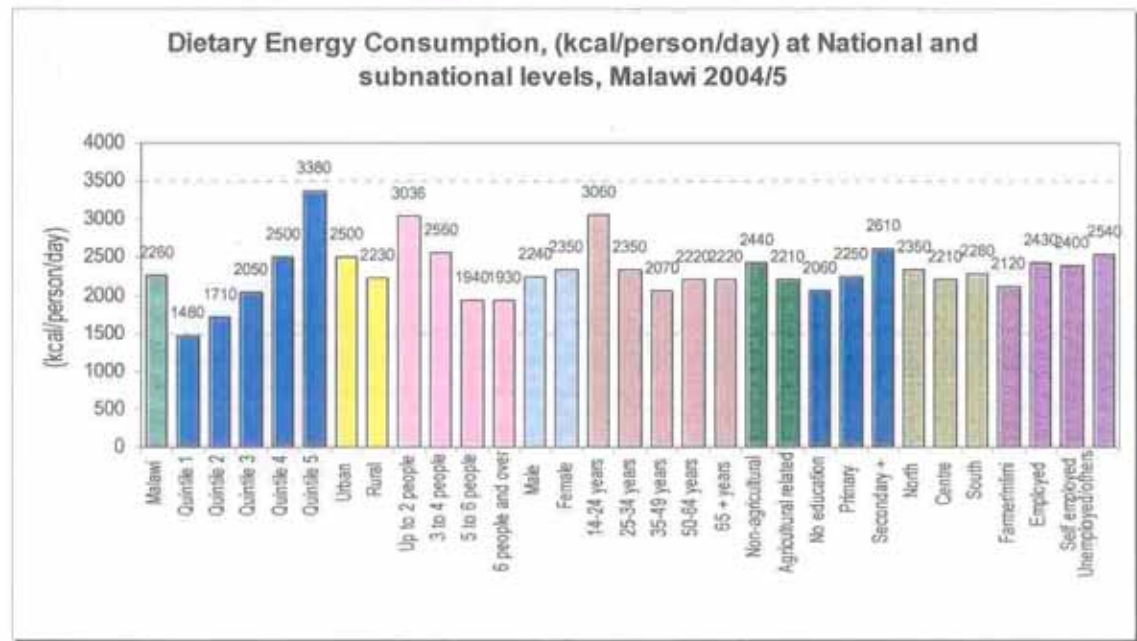
The daily average dietary energy consumption (DEC) in Malawi in 2004 was 2260 kcal/person. The dietary energy consumption levels increased with increasing income levels ranging from 1480 kcal/person/day for people in the lowest income quintile to 3380 kcal/person/day for those in the highest income quintile. This could be attributed to the availability of income to purchase more food (for the highest quintile.) DEC for people in the first income quintile was respectively 1480 kcal/person/day, which was lower than the MDER of 1628 kcal/person/day.

The urban population had a higher daily average Dietary Energy Consumption (DEC) of 2500

kcal/person/day as compared to 2230 kcal/person/day of the rural consumption. This could be because most of the rural people have lower income levels as compared to their urban counterparts and again this could also be attributed to the diversity of foods the urban people eat as compared to the rural masses.

Smaller households i.e. those with only two members or less had a higher dietary energy consumption of 3036 kcal/person/day, as the household size increased the DEC decreased with the increase in household members. Households with more than 6 members had a lower average DEC of 1930 kcal/person/day; this is due to spreading the little that is available amongst a larger group. (Figure below).

Figure 7: Dietary Energy Consumption, (kcal/person/day)



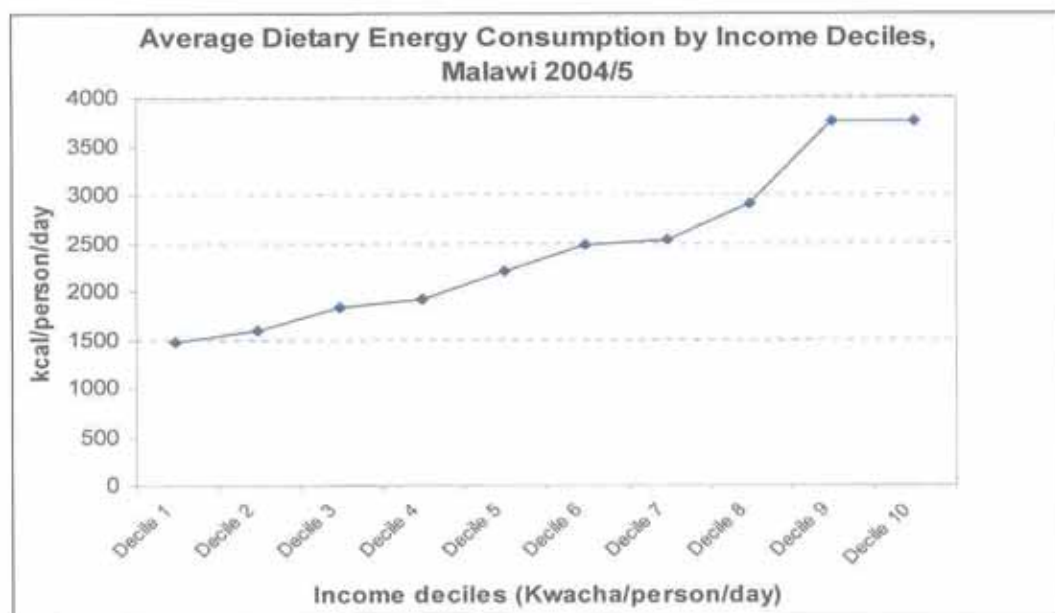
There were marginal differences in DEC between households headed by male and female, This could be attributed to the fact that most female headed households prioritised their expenditure to food items than most male headed households. However the survey showed that all gender categories were food secure by the survey period. However, population in households with heads in a non agricultural activity had higher energy consumption than in those with head in the agricultural activity.

There was a minimal variation in DEC across regions; however, the northern region of Malawi registered the highest Dietary Energy Consumption of 2350 kcal/person/day whilst the central part of Malawi had the lowest Dietary Energy Consumption of 2210 kcal/person/day. The disparity could be because of the fact that in the Northern and southern parts of the country there is some Dietary diversification i.e. there is a wide variety of crops e.g. cassava, rice, maize, bananas etc which they use as food whilst the central part of Malawi mostly people rely on maize

as the staple food crop. There were no major variations in Dietary Energy Consumption in the employment sector however those people that were farmers had almost a breakeven Dietary Energy Consumption of 2120 kcal/person/day, (Figure below).

The figure below shows the increasing trend of DEC with increase in household income.

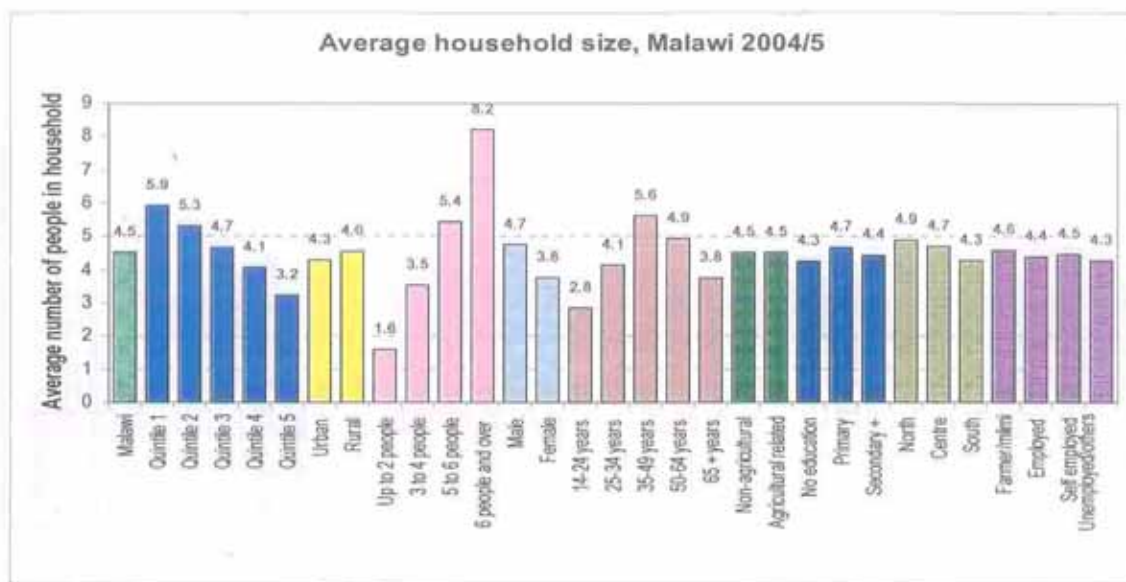
Figure 8: Average dietary energy consumption (kcal/person/day)



Effect of Household size

The average household size for the country was estimated as 4.5. The low-income population were in households having on average a high number of members (5.9) resulting to a lower DEC and a high level of food deprivation. Level of food deprivation in Malawi was closely related to factors of large household size and low income. Households in the highest income group had an average low household size of 3.2 and also a low level of food deprivation.

Figure 9: Household size (average number of people in the household) at national and sub national level

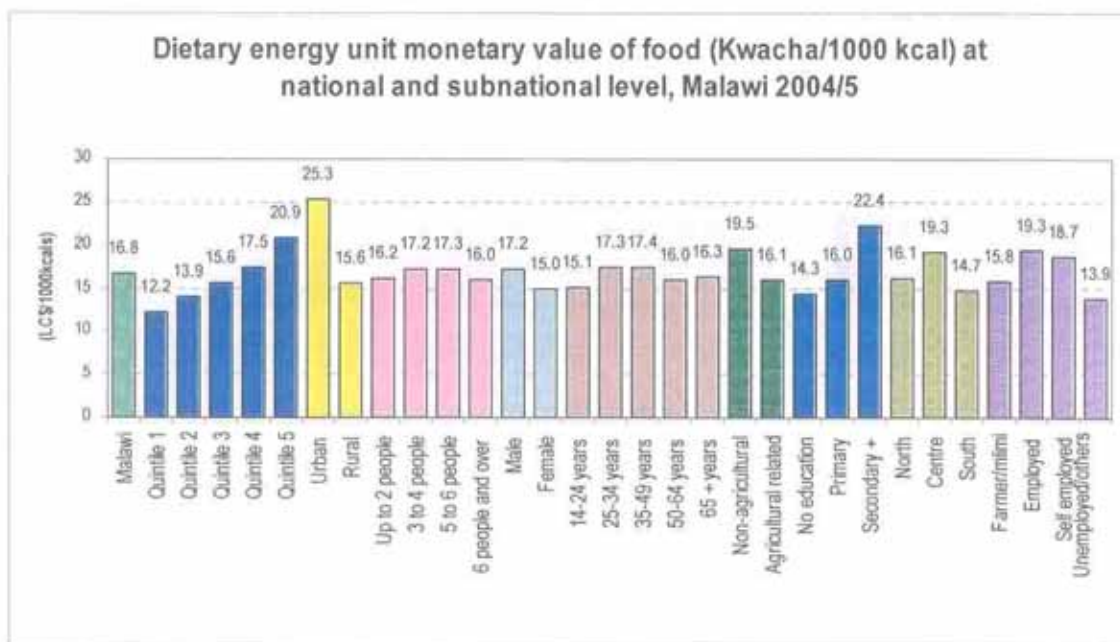


Male headed households had a slightly higher number of household members of 4.7 members whilst the female headed households had an average number of 3.8 members hence slightly higher DEC as compared to male headed households.

2. Dietary energy unit value

It costs on average MK16.8 to acquire 1000 kcal at national level. The dietary energy unit value (excluding preparation costs such as cost of energy needed to bring the food from as purchased to ready to eat) vary among the different income quintile groups. The values range from MK12.2 \$ for the low-income groups usually acquiring low price food to MK 20.9 for the high-income quintile population. The dietary energy unit values were higher in urban areas, in male headed households, in those households whose heads attained secondary education and higher, and employed. This could be due to higher income levels and also because most populations in rural areas and in female headed households obtain food from own production hence less money spent on food.

Figure 10: Dietary Energy unit monetary value of food (Kwacha/1000 kals)



3. Monetary value of food consumed and Engel ratio

On average, it costs MK37.9 per person per day to a Malawian to acquire a food basket corresponding to the national consumption of 2260 kcal.

The population group in the higher income quintiles of 4 and 5 spent a higher amount on their calories; their average cost is MK43.7 and MK81.9 respectively, while those in the lower quintiles obtain their energy at a lower cost of MK18.0 which is below the national average.

At regional level, the populations residing in urban areas have a higher monetary value of Mk63 as compared to the rural areas with a cost of MK 34/person/day. This is due to the fact that the rural consumption is mostly from own production.

Households whose heads attained higher education levels, are employed, are aged below 34 years, engaged in non-agriculture related activities and households less than 2 people, spend more than the national average. However, there were minor differences observed between male-headed and female headed households.

Differences were found mainly between the income quintiles with the spending the cost of MK18.0 while those of the highest income group paying an amount of MK81.9. Marginal differences were noted among regions and age of the household head. However, population of smaller households, households living in urban areas, households with head with high education and employed spent more in acquiring their food (Figure below). This could be associated with higher income levels. Figure 11: Monetary value of food consumed (kwacha/person/day) at national and sub national level.

Figure 12 : Monetary value of food consumed (kwacha/person/day)

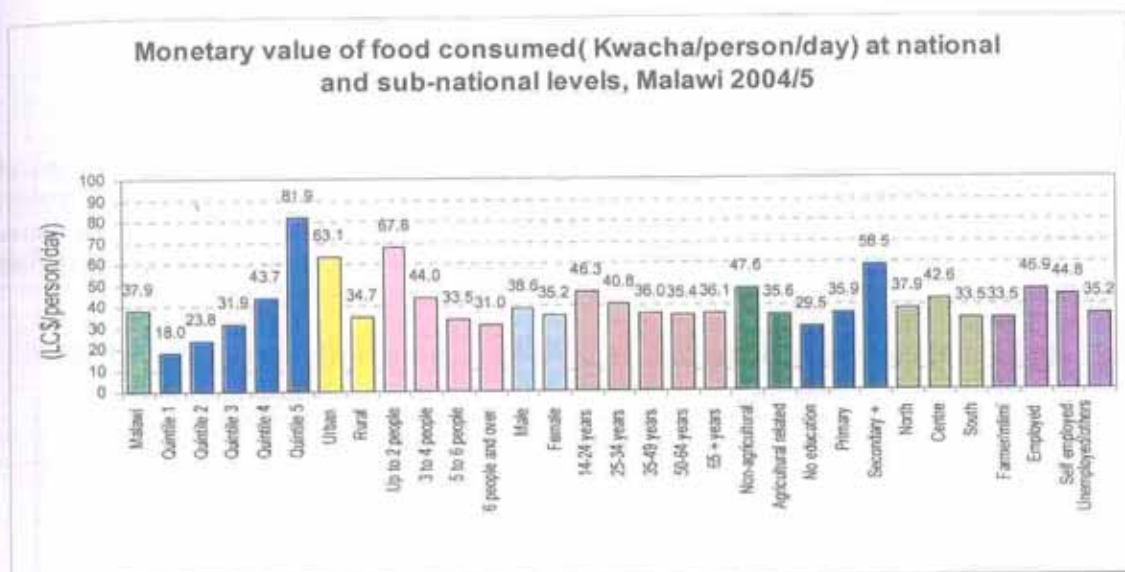
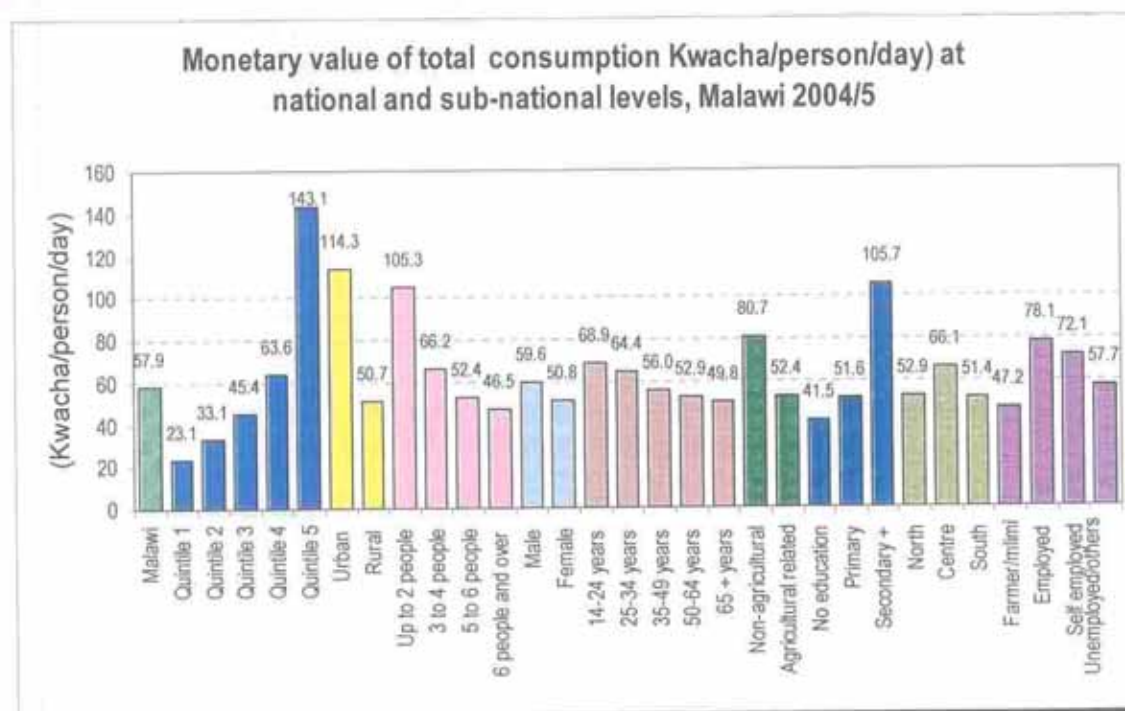


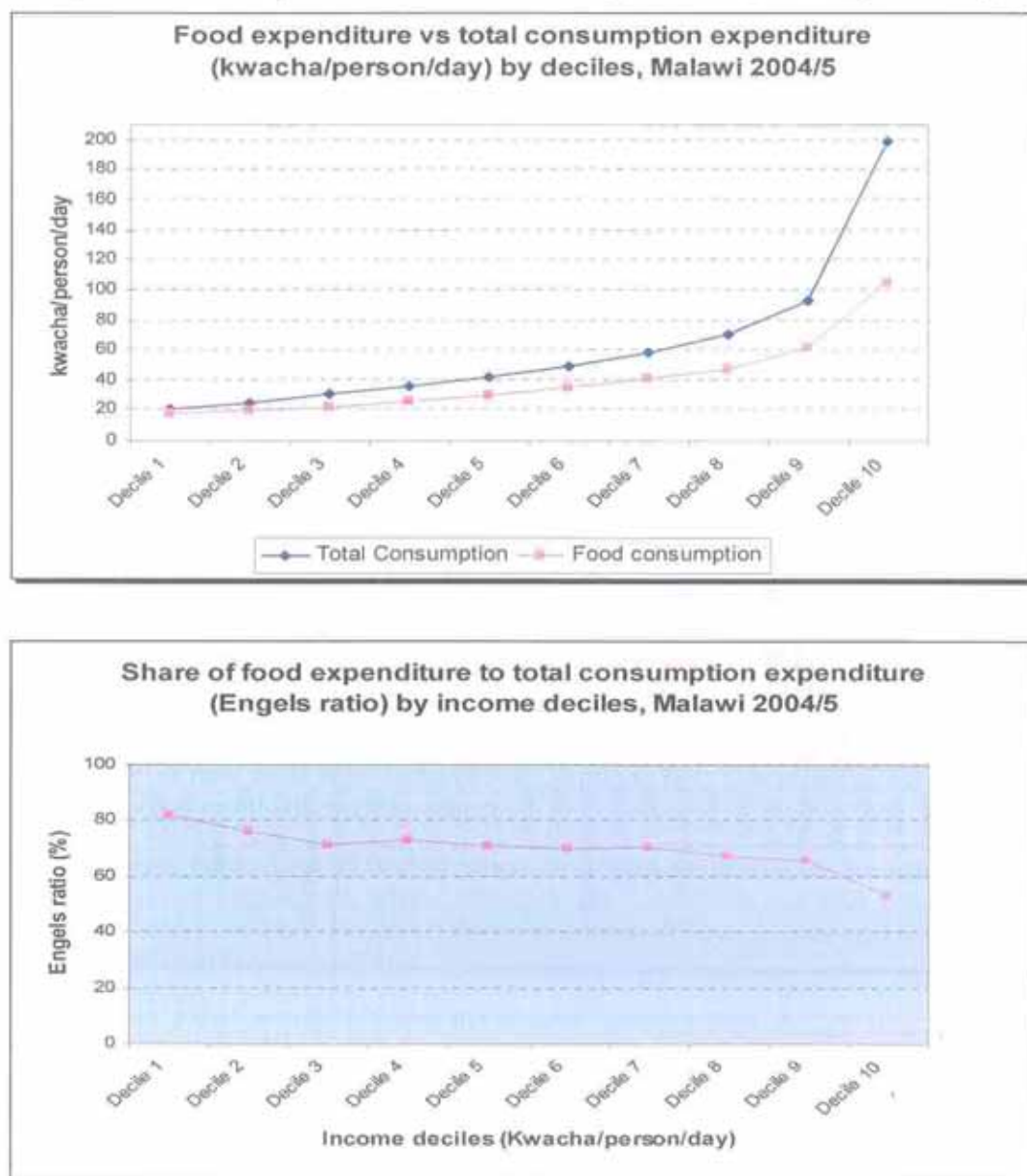
Figure 13: Monetary value to total consumption (kwacha/person/day)



3.1 Engel's ratio

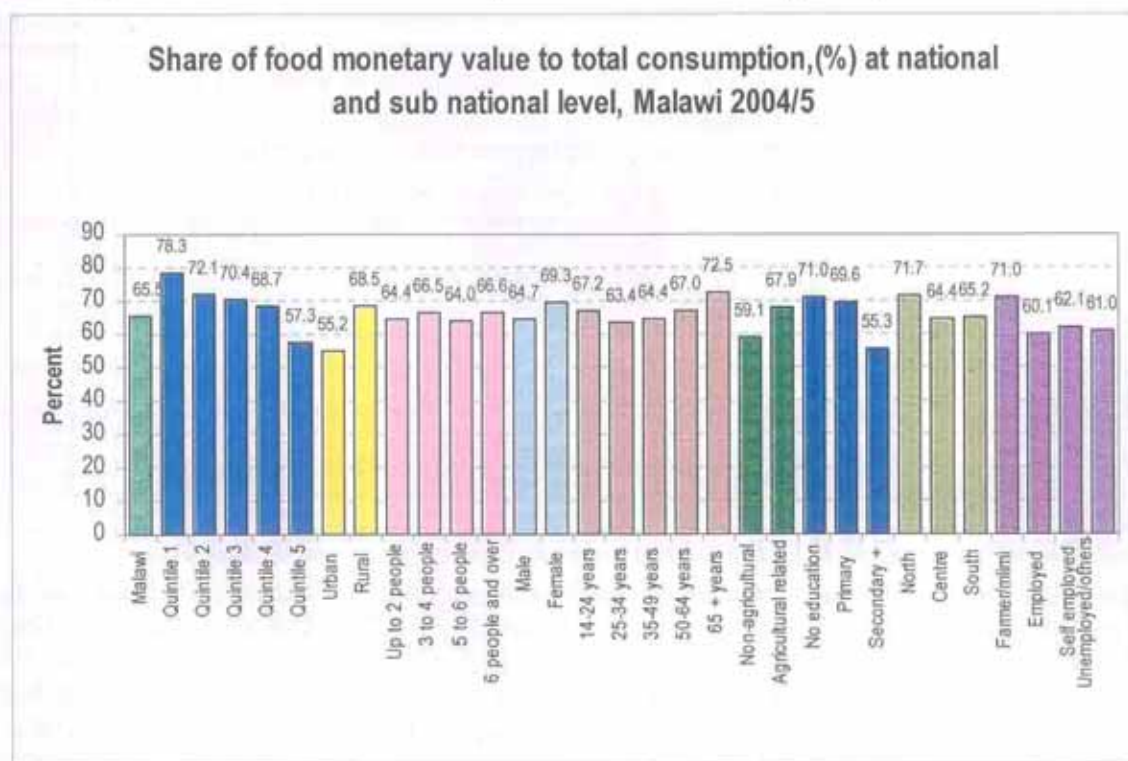
The share of food expenditure to total consumption known as Engel ratio is an indicator of consumption pattern, and usually is closely related to income. Figure 10 and 11 below, however, shows that the Engel ratio did not vary much for deciles from 1 to 8, oscillating about the value of 53 percent with standard deviation of 1.1 percentage points. Only for the top, 10th quintile did it go far below 50 percent.

Figure 14: Food expenditure and Total food expenditure, (Kwacha/person/day)



On average, the share of food monetary value to total consumption is 65.5% but the share is higher in populations in rural areas, in households within lowest levels of income, in female headed households and in households whose heads are involved in agriculture (including farmers), had no education and the age is more than 65 years. This is because their main concern is food and not much on luxury goods (refers to figure below).

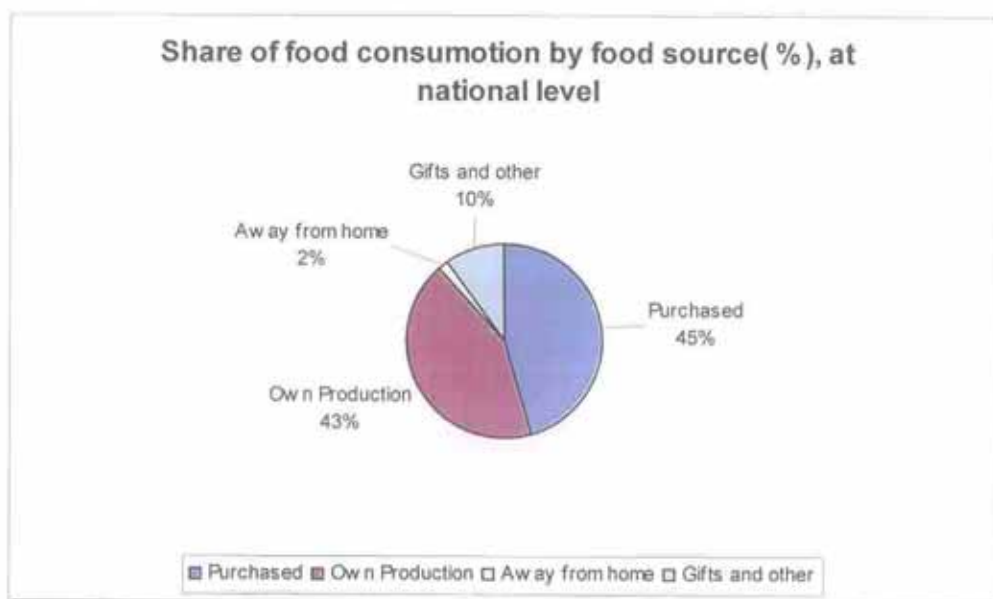
Figure 15: Share of food monetary value in total monetary value, Malawi 2004/5



4. Share of food consumption by food source

Food consumption by households is acquired and consumed mainly through purchases, own production or given as payments in kind or gifts, in addition to some consumption outside home in restaurants or from street vendors, etc. At national level, 45 percent of the population acquired most of their food consumption from purchases and 43 percent from their own production. The latter was considered significant and probably due to the high rural population whereby people resort to have their own food production to meet their food needs. The shares of food consumption from the other two sources are marginal, particularly of the away from home consumption, 10 percent and 2 percent respectively.

Figure 16: Share of food expenditure by food source (%), Malawi 2004/5

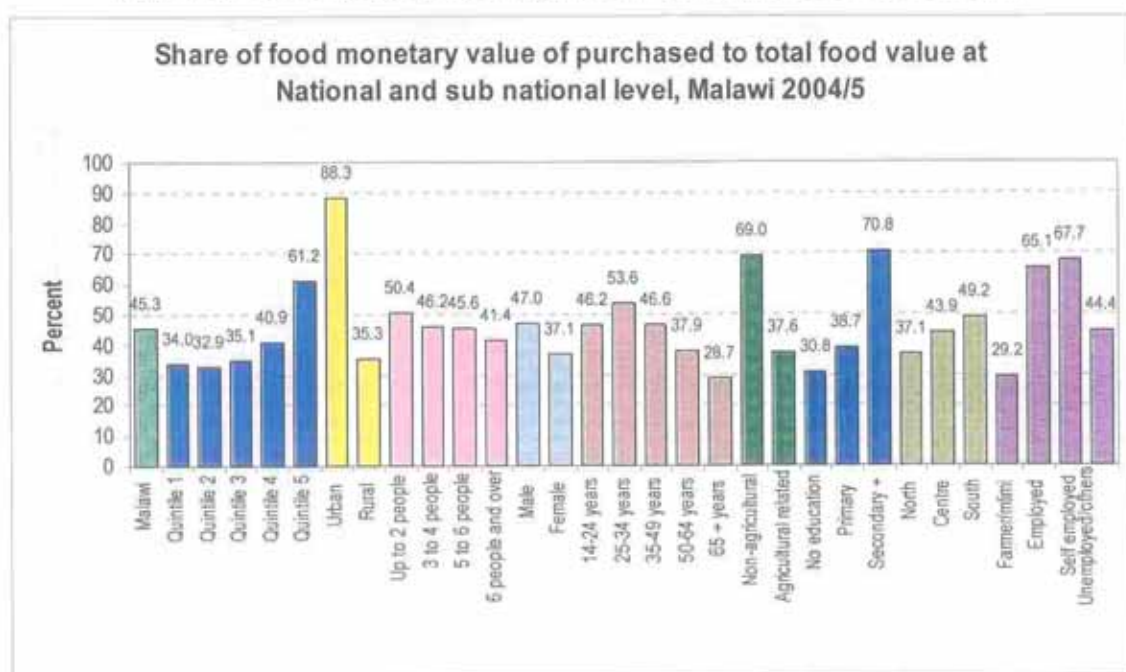


The population of urban regions acquired a bulk of their food from purchases while those of the rural areas derived about 45 percent of their food consumption from their own production. Own production remains an important food source for most of the population groupings, even the population of the highest quintile. Food eaten away from home was almost insignificant with of the entire population among most population groupings. The structure of food dietary energy consumption followed almost the same pattern as the structure of food consumption expenditure and is depicted in the sections below.

4.1 Share of food monetary value from purchase to total food value

The national average of share of food monetary value from purchase is 45.3 %. There are higher percentages in households with higher income levels, with less than 2 people, living in urban areas, in male headed households and those households whose heads are employed and those that are involved in non-agriculture related activities. This could be due to high income levels and low dependability on own production.(refer to table 14 below).

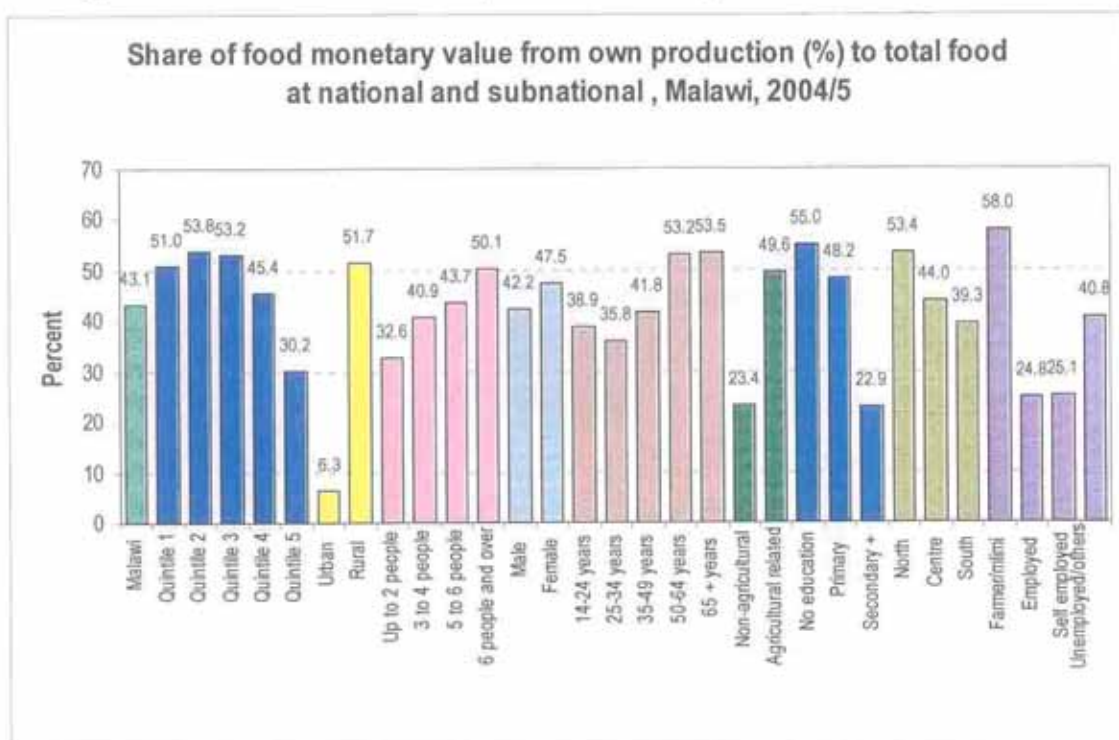
Figure 17: Share of food monetary value of purchased to total food value



4.2 Share of food monetary value from own production to total food value

The national average share of food monetary value from own production to total food value was 43.1%. This share was higher in the first three quintiles, in population residing in rural areas, in households whose heads are females, in households whose heads are more than 50 years old, have no education and are engaged in agriculture related activities. This could be attributed to the fact that they rely more on own production as a source of food, coupled with low income status (refer to figure 11 below).

Figure 18: Share of food monetary value from own production to total food value

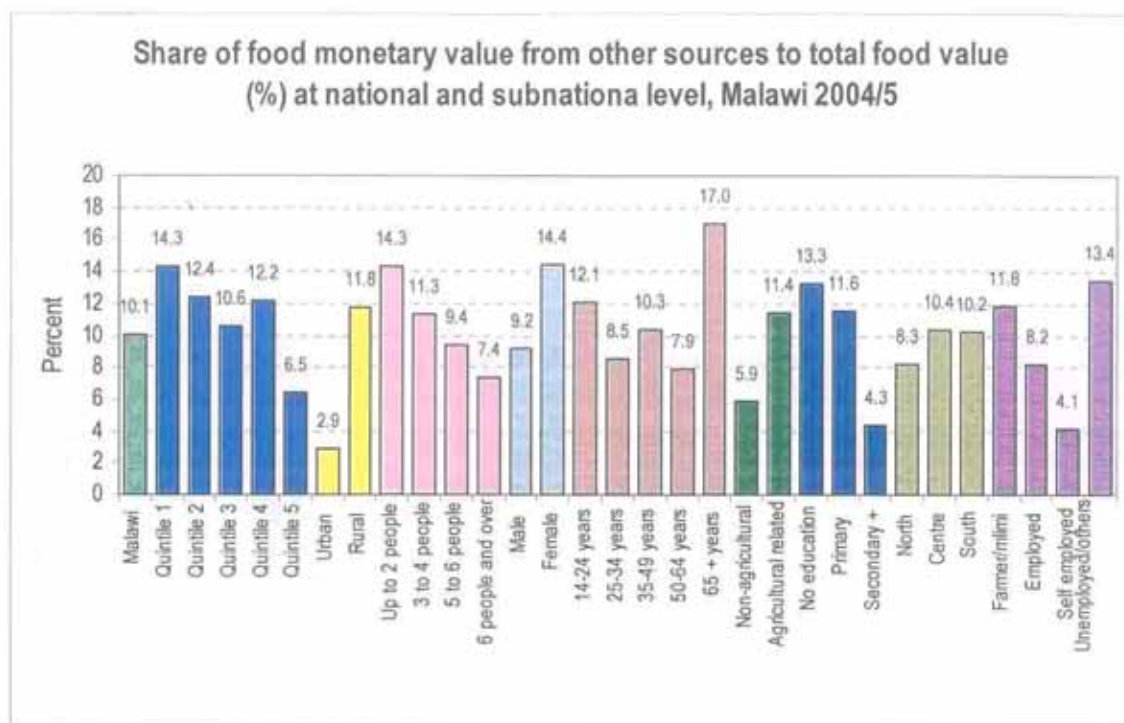


4.3 Share of monetary value of food from other sources

On average the nation's share of monetary value of food from other sources is 10.1%. There is higher percentage in the lower quintiles, in rural populations, in female headed households, in the households whose heads are more than 65 years old, are not employed, have no education, are engaged in agriculture related activities.

This could be because these households are vulnerable due to their low income status hence they sometimes rely on free food distributions and even gifts from relatives. For more details refer to table 13 below.

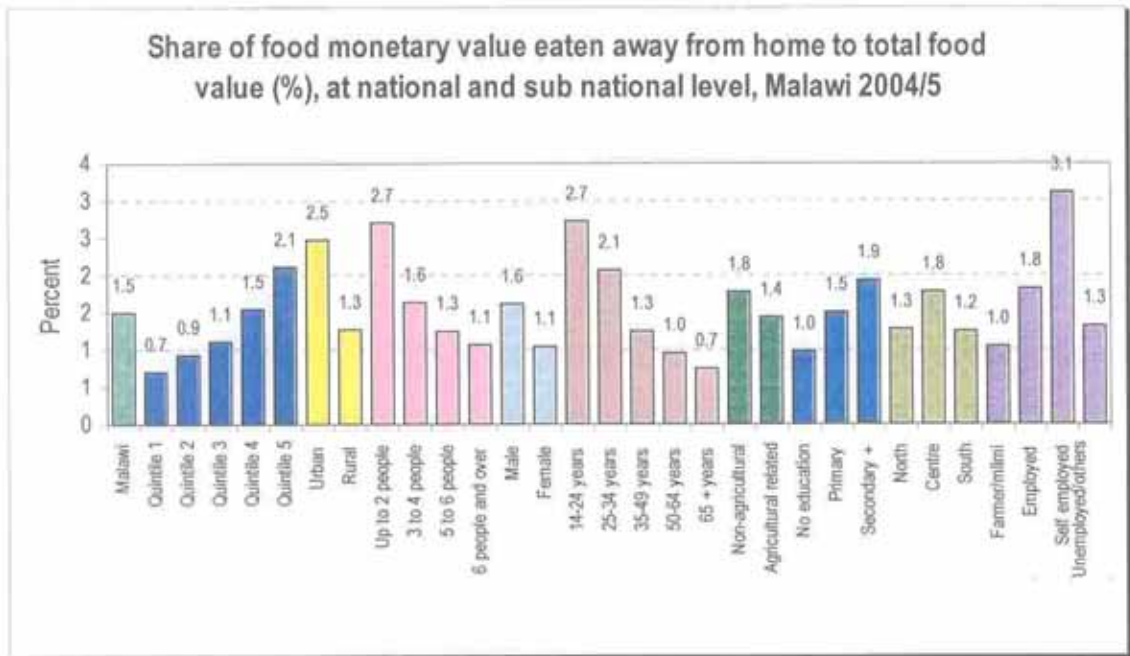
Figure 19: Share of monetary value from other sources to total food value



4.3 Share of food monetary value of food eaten away from home

The average share of food monetary value of food eaten away from home is 1.5% but populations in quintile 5, in urban areas, in male headed households, and in those households whose heads are aged below 34 years, have attained higher education, are employed and are working in non-agriculture related activities have higher percentage of food monetary value of food eaten away. This could be because their income status is higher; therefore they can afford to buy more expensive foods such as those prepared at restaurants.

Figure 20: Share of food monetary value eaten away from home

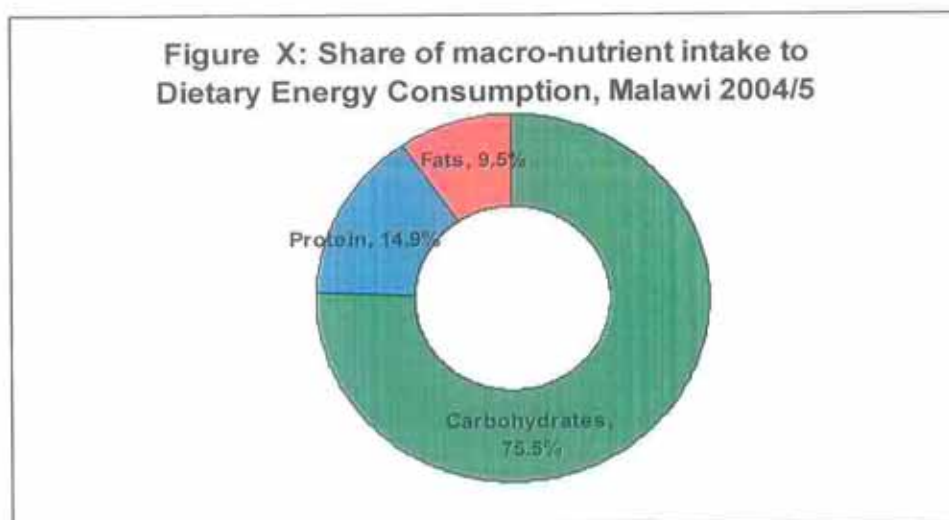


The shares of purchased food in terms of dietary energy were higher in terms of monetary value. This could be due to the fact that food from own production were mostly vegetables and fruits which are poorer in calories than cereals which are mostly purchased, especially in urban areas, and constitute a considerable part of population's diet (see Figure 20 below).

(E) Dietary Diversity

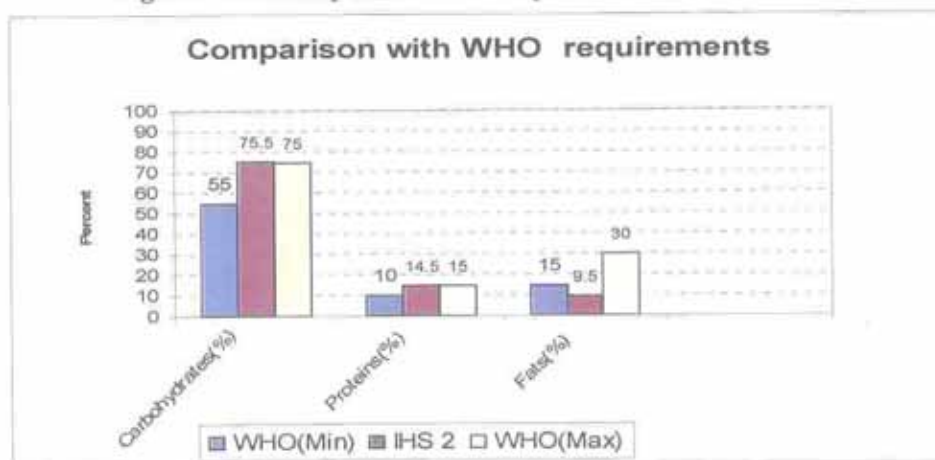
The results shows that the average dietary energy consumption (DEC) of Malawian households is mostly carbohydrates, 76 percent, followed by proteins 15 percent and fats 10 percent, as shown in the (Figure below).

Figure 21: Share of macronutrient intake to dietary energy consumption, Malawi 2004/5

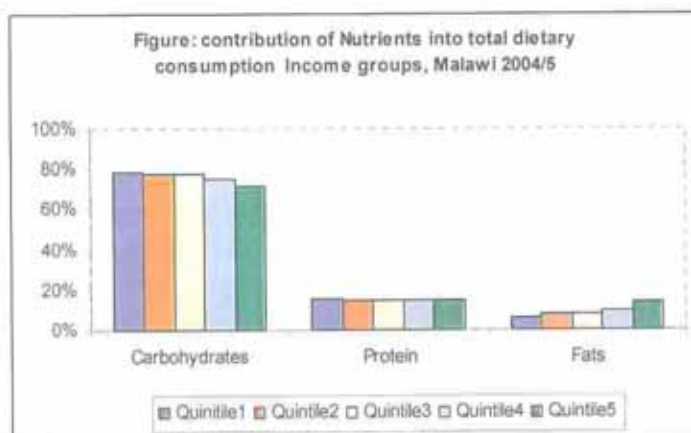


Compared with the norms recommended by WHO for a balanced diet to contain 10-15 percent of proteins, 15-30 percent of fat and finally 55-75 percent of carbohydrates. The findings show that the Malawian diet is slightly biased towards carbohydrates, being at the maximum of the requirements 75 percent, and at the minimum of the requirements on proteins and fats,

Figure 22: Dietary intake in Comparison with WHO standards

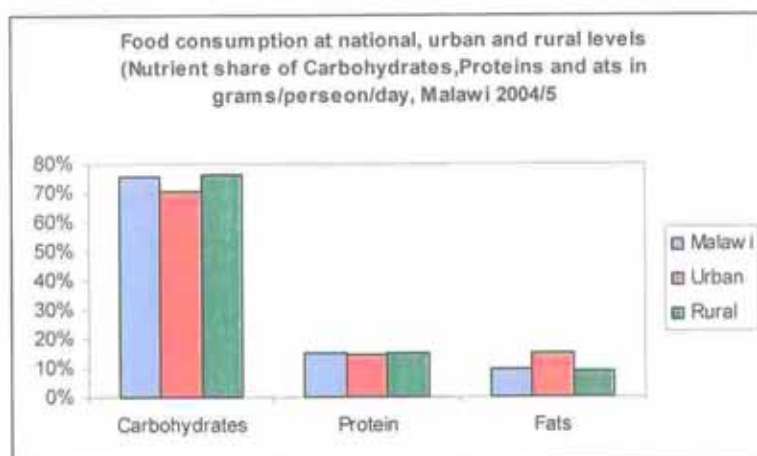


Income disparities



Looking closely at the dietary balance distribution across the household background characteristics, it can be shown that across the income quintiles the nutrient contributions are the same however as you move to the highest income quintile 5, the contribution of fats intake goes up, while that of carbohydrates goes down, as shown in the figure to the left.

Geographical disparities

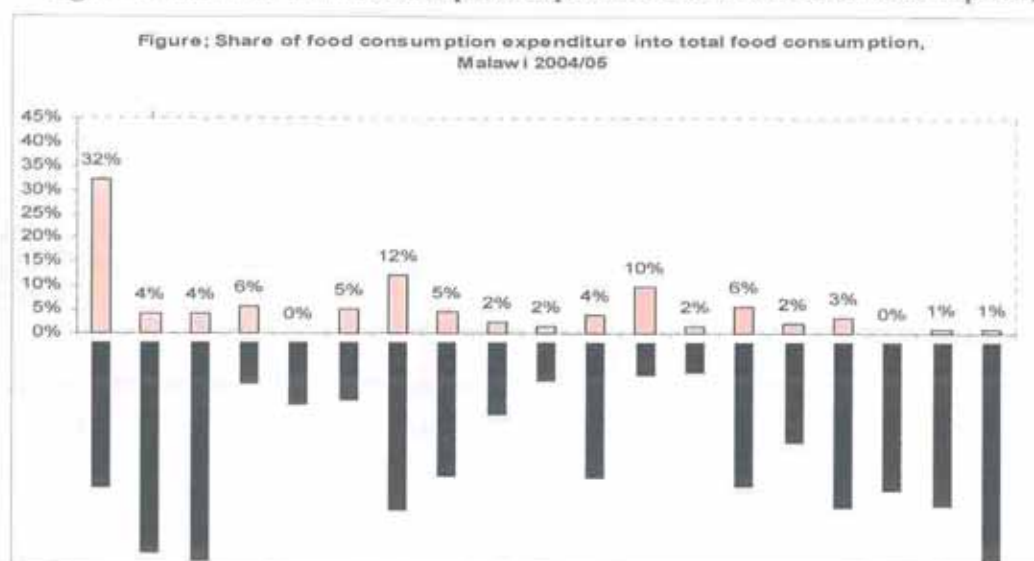


The geographical disparity of nutrient intake into food dietary consumption shows that, the rural consumes more carbohydrates (76 percent) than the urban areas (71 percent), almost similar on proteins, however on fat intake the urban areas are higher, 15 percent compared to rural 9 percent, as shown in table to the left.

The percentage contribution of dietary energy consumption by main food items groups are shown in the figure below, at the national level, Cereals and cereals products constituted the main dietary energy supply for the average Malawian with a contribution of 48 percent to total dietary consumption. This was followed by oils and fats from vegetable and plant origin with a 9.5 percent share contribution to total dietary, roots and tubers contributed 8.3 percent, followed by pulses which contributed 8 percent to total dietary consumption. The least food commodity groups consumed were Non alcoholic beverages and prepared foods and spices contributing less

than 1 percent. Oils and fats of animal origin and tree nuts were not consumed by the households.

Figure 23: Share of food consumption expenditure into total food consumption,



The contribution of the food commodity groups to the nutrient share and the total dietary consumption is also shown in the table below. Cereals remain the major source of dietary energy consumption contributing to protein, carbohydrates share and fats share. In the protein consumption, Cereals are the main source of protein contributing 35.4 percent of the proteins, Fish and fish products are the second source of protein in Malawi contributing a share of 21 percent; this is followed by oil crops contributing a share of 11.4 percent protein.

For the carbohydrates intake, cereals and cereal products remain to be the major source contributing a share of 63 percent, followed by roots and tubers such as cassava and potatoes, contributing 11.8 percent and sugars 8.3 percent.

For fats the dietary contribution is mainly from oils and fats of vegetable origin 50 percent , Oil crops such as groundnuts, 17 percent and cereals and products 11 percent.

Table 1: Share of each nutrient in total dietary energy consumption by food commodity groups

<i>Region: Africa</i>	<i>Country: Malawi</i>	<i>Year: 2004/05</i>		
Food commodity group	Share of dietary energy consumption in Total Energy Consumption (%)	Share of protein consumption in Total Protein Consumption (%)	Share of carbohydrates Consumption in Total Carbohydrates Consumption (%)	Share of Fats Consumption in Total Fats Consumption (%)
CEREALS AND PRODUCTS	48.9	35.4	63.0	11.0
ROOTS AND TUBERS AND PRODUCTS	8.3	2.1	11.8	0.8
SUGARS AND SYRUPS AND PRODUCTS	5.5	0.0	8.3	0.0
PULSES	7.7	15.1	7.8	1.5
TREE NUTS	0.0	0.0	0.0	0.0
OIL CROPS	6.7	11.4	2.3	19.0
VEGETABLES AND PRODUCTS	1.7	3.3	1.5	0.5
FRUITS AND PRODUCTS	1.7	0.7	1.9	1.2
STIMULANTS	1.3	2.7	1.1	0.5
SPICES	0.1	0.4	0.1	0.1
ALCOHOLIC BEVERAGES	1.7	1.1	1.7	0.1
MEAT	1.9	3.9	0.0	7.1
EGGS	0.2	0.5	0.0	0.7
FISH AND FISH PRODUCTS	3.8	21.4	0.0	5.1
MILK AND CHEESE	0.5	0.8	0.2	1.5
OILS AND FATS (vegetable oils)	9.5	0.0	0.0	50.2
OILS AND FATS (animal fats)	0.0	0.0	0.0	0.1
NON ALCOHOLIC BEVERAGES	0.1	0.0	0.2	0.0
MISCELLANEOUS AND PREPARED FOOD	0.4	1.0	0.2	0.8
Integrated Household Survey 2004/05				

The unit cost of nutrients by food commodity groups shows large disparities within and between nutrients. It should be noted the unusually large unit values are due to the fact that some items are low in corresponding nutrient values. The cheapest dietary energy was provided by cereals, sugar and sugar products and vegetable oils (Table 3). The cheapest protein was derived from cereals, cheapest carbohydrates from sugar while the cheapest fat from vegetables oils. High nutrient values correspond to food groups that are not sources of that nutrient.

Table 2: Nutrient costs by food commodity groups

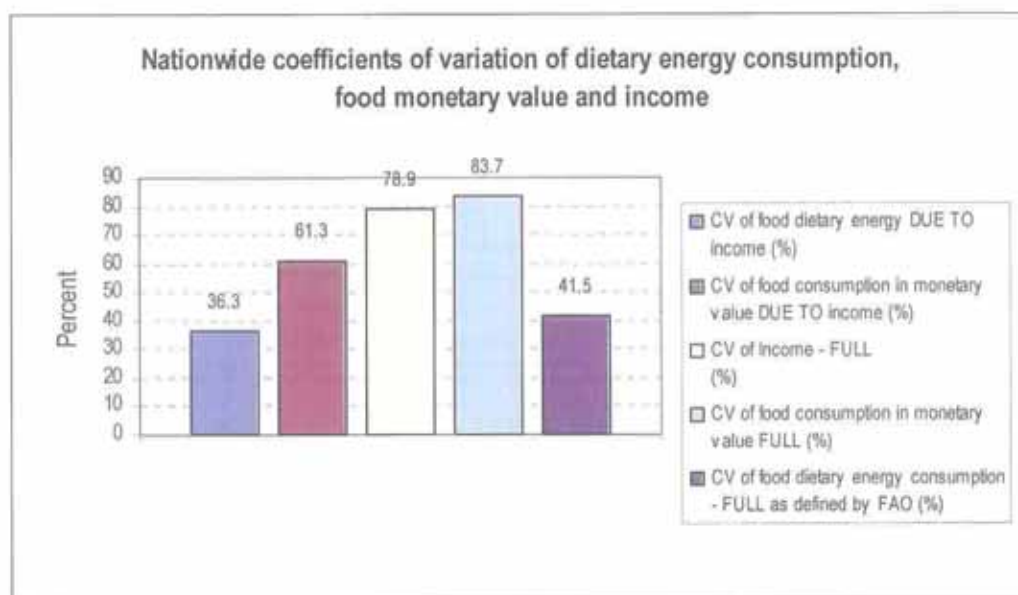
Region: Africa		Country: Malawi		Year: 2004/05	
Food commodity group	Average Dietary Energy Unit Value (LC\$/1000kcal)	Proteins Unit Value (LC\$/100g)	Carbohydrates Unit Value (LC\$/100g)	Fats Unit Value (LC\$/100g)	
CEREALS AND PRODUCTS	11.13	46.65	5.19	235.43	
ROOTS AND TUBERS AND PRODUCTS	8.12	99.41	3.48	418.31	
SUGARS AND SYRUPS AND PRODUCTS	13.01	24897.28	5.24	32707.78	
PULSES	12.47	19.37	7.39	309.36	
TREE NUTS	0.00	0.00	0.00	0.00	
OIL CROPS	12.52	22.49	22.15	21.03	
VEGETABLES AND PRODUCTS	122.31	183.56	83.53	1816.60	
FRUITS AND PRODUCTS	45.86	307.23	23.98	298.88	
STIMULANTS	30.18	42.31	20.44	370.19	
SPICES	181.74	194.27	142.09	1835.02	
ALCOHOLIC BEVERAGES	37.72	169.00	22.40	5024.56	
MEAT	86.73	125.16	260242.68	108.02	
EGGS	136.16	158.90	2745.36	192.25	
FISH AND FISH PRODUCTS	25.32	13.58	0.00	89.73	
MILK AND CHEESE	69.33	143.91	102.80	116.35	
OILS AND FATS (vegetable oils)	5.82	0.00	0.00	5.24	
OILS AND FATS (animal fats)	61.17	0.00	2861.67	55.53	
NON ALCOHOLIC BEVERAGES	121.79	0.00	48.72	0.00	
MISCELLANEOUS AND PREPARED FOOD	41.06	50.47	56.75	97.13	

Malawi, Integrated Household survey, 2004/5

(f) Food and income inequality

Inequality in access to food is estimated by FAO through the Coefficient of Variation (CV) of dietary energy consumption. This CV results of two components to capture on one side the variation of DEC owing to income and on the other side the variation of DEC owing to energy requirement. The later has been set to 20 percent. The CV of DEC due to income was 36.3 percent resulting to an overall CV of DEC of 41.5 percent including CV due to requirement. The CV of food consumption in monetary value due to income (61.3 percent) is higher than that in dietary energy (36.3 percent) as the former includes the variations of food prices. CV of income due to income commonly called CV income FULL was relatively high (78.9 percent) than the CV of DEC.

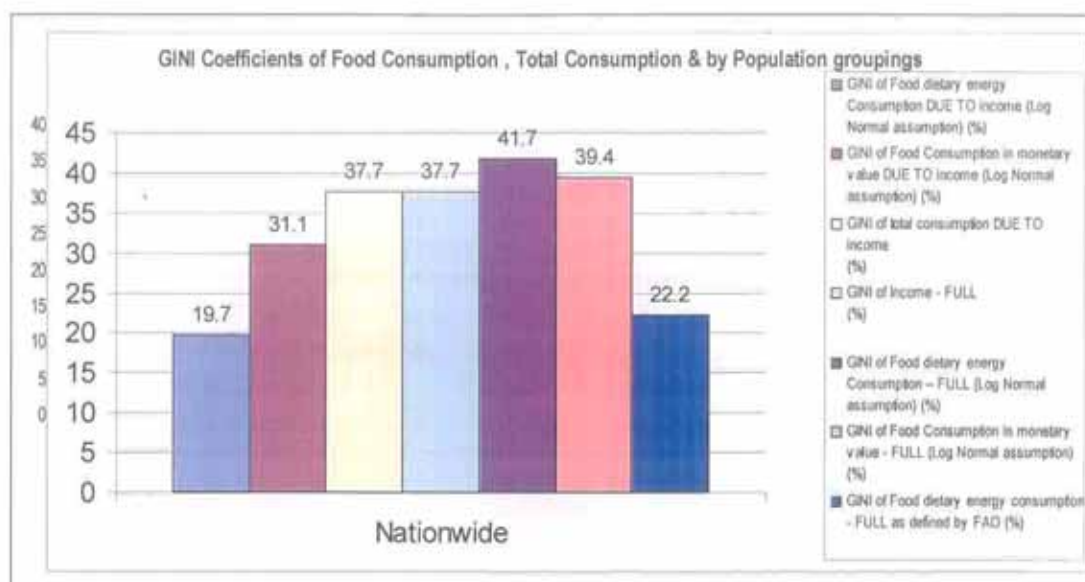
Figure 24: Nationwide coefficients of variation of DEC, monetary value and income



1. Inequality in access to food consumption due to income at sub national levels

Inequality in access to food as measured by the CV of DEC due to income was at 19.7 percent nationwide whilst the Inequality of food consumption (monetary value) due to income (%) was at 31.1 percent. Lower inequalities were observed in households with 5 to 6 family members and those with 3 to 4 family members.

Figure 25: GINI Coefficients of food consumption



2. Dispersion (inequality) ratios

The dispersion inequality of a variable measures the disparity between the lowest and highest quintile (80/20). It is estimated for the food consumption in both monetary and dietary energy values, consumption and income. At national level, the dispersion ratio for food consumption in dietary energy was 2.7, in monetary value, 4.5, in total consumption 6.2 and 6.2 in income. Dietary energy consumption has variability less than food expenditure as the latter includes additional variability of food prices. Total household consumption includes variability of all commodities by types and prices. Variations in income include many components of socio economic factors. Dispersion ratios estimated for various sub national levels for each of the four components are given in the annex tables.

Figure 26: Inequality as measured by Quintile dispersion ratio, National

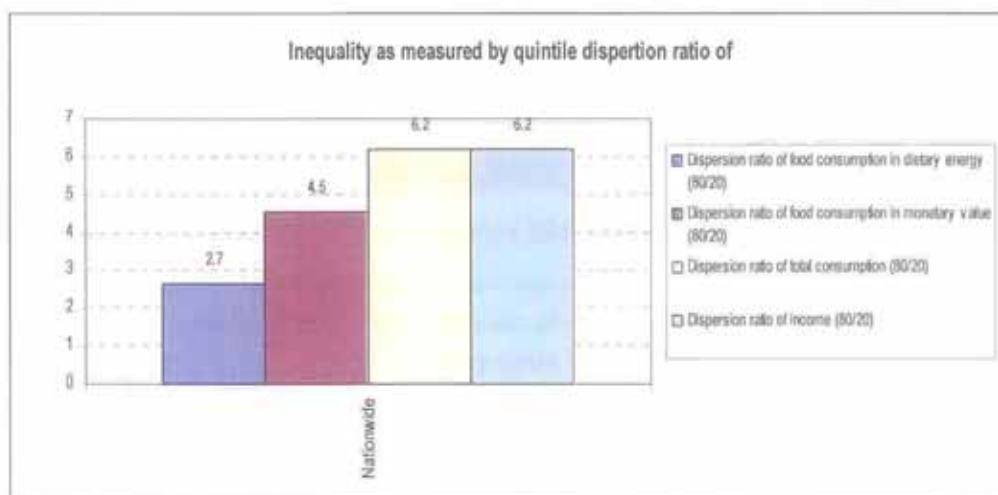


Figure 27: Inequality as measured by Quintile dispersion ratio, Male and Female

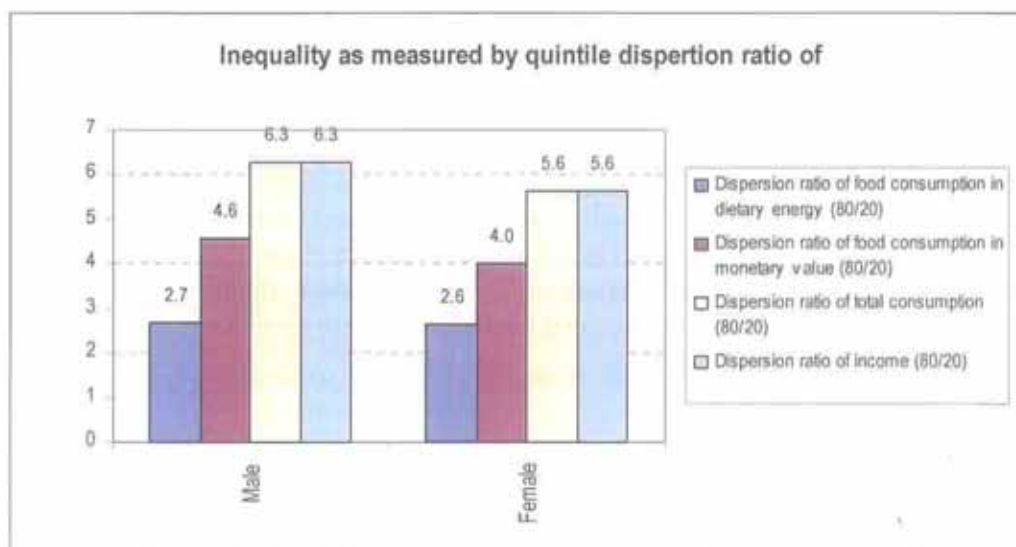
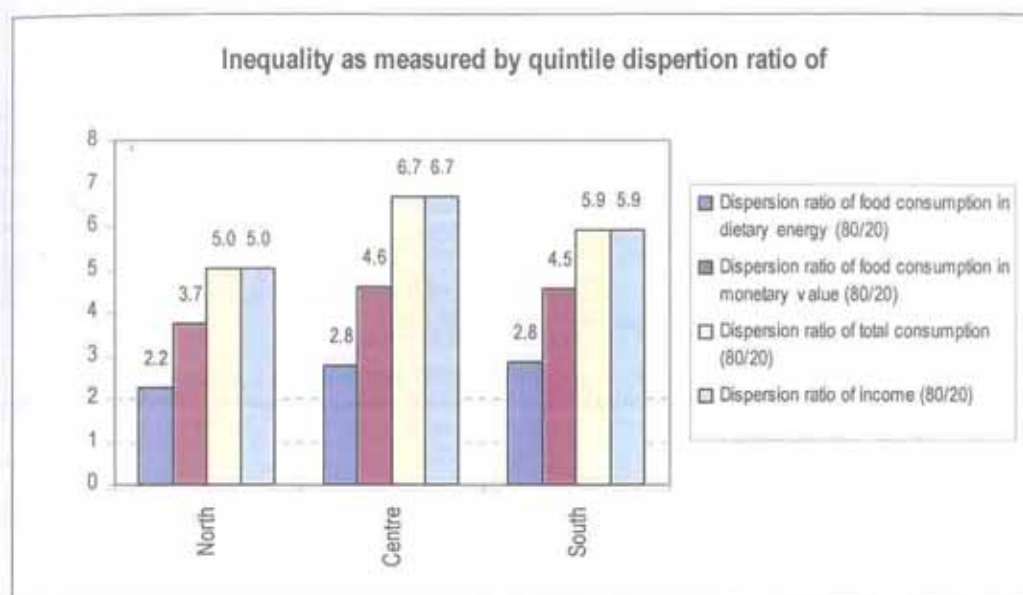


Figure 28: Inequality as measured by Quintile dispersion ratio, by regions



4. Income Demand Elasticity of Food Consumption

The income demand elasticity of food consumption in terms of dietary energy and expenditure have been computed as a function of income as the latter is high correlated to food consumption. The income elasticity of food dietary energy is less than that of food consumption expenditure. Elasticity values of DEC decreased with increasing income deciles. The positive responses of 1.1 percent in dietary energy consumption of the lowest income decile shows that increase of 1 percent in income is not substantially translated into improvements in the nutrition of the poor.

However, the observation with relation to food expenditure is different, where the lowest decile had elasticity 5.7 percent far above one showing very high sensitiveness with increase in income. Population will respond in their food expenditure purchasing expensive food products having almost the same dietary energy. The demand elasticity of the food share is stable around the value of 0.9.

Table 3: Income Demand elasticity with respect to Income and population groupings, Malawi 2004/5

Categories and Groupings	Income deciles	Income (LC\$/person/day)	Demand elasticity of food dietary energy consumption (DEC) respect to income (%)	Demand elasticity of food consumption in monetary value (FMV) respect to income (%)	Demand elasticity of Engle ratio respect to income (%)
Income level					
Decile	1	20.65	1.1	5.7	0.9
Decile	2	24.86	0.9	2.8	0.9
Decile	3	30.58	0.7	1.8	0.9
Decile	4	35.52	0.7	1.4	0.9
Decile	5	41.96	0.6	1.1	0.9
Decile	6	48.92	0.6	1.0	0.8
Decile	7	57.65	0.5	0.8	0.8
Decile	8	70.01	0.5	0.7	0.8
Decile	9	93.12	0.4	0.6	0.8
Decile	10	199.11	0.3	0.4	0.8

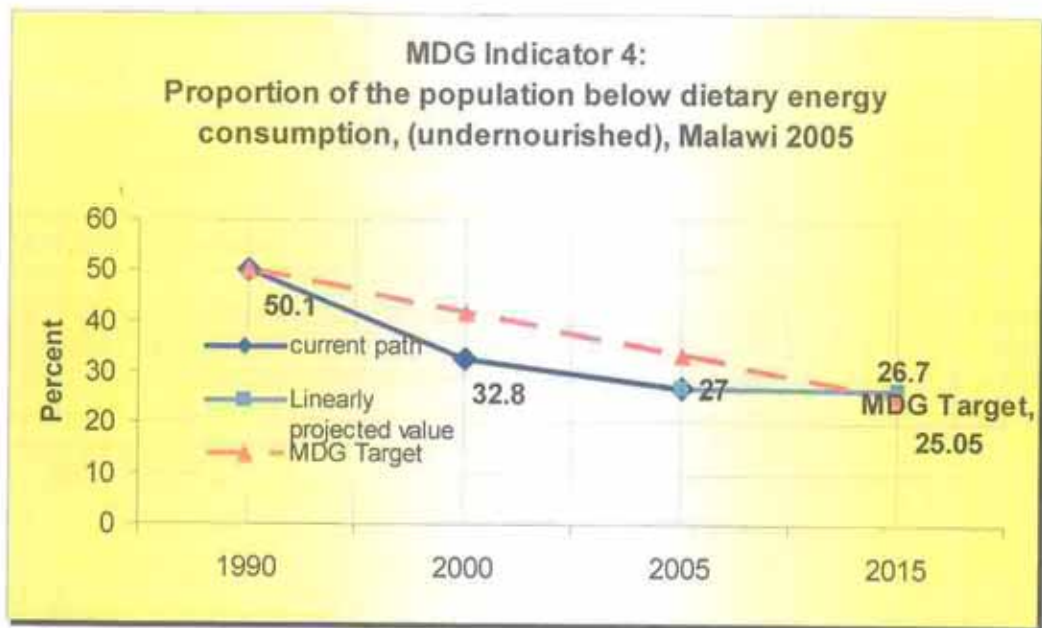
(G) Progress towards the WFS and MDG

The estimate of the MDG indicator 5 from the IHS2 was 27 percent, this is in tandem with the trend of the previous estimates from FAO, which were 50.5 percent in 1990, and 32.8 percent in 2000-

Accordingly to the indicator Malawi has performed very well, the target for MDG was to reduce the 1990 value by half, to 25 percent. If the trend of reduction improves this is estimated at 1.04 percent annually. By 2015 Malawi would have reached 26.7 percent. This is slightly above the MDG target of 25.05 percent by 1 percent.

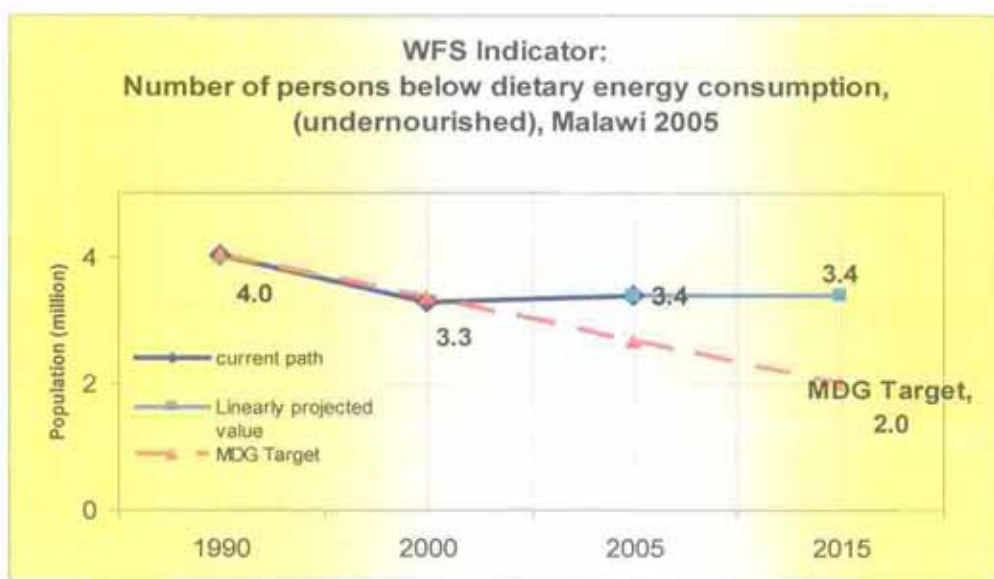
It should be noted that past estimated by the FAO estimates were derived using available food supply data while IHS uses household consumption data. Thus these figures should to be taken with caution, particularly that Georgia has a decreasing population growth and it is important that the FSSM be applied to previous IHS to analyse the trend of the prevalence of under nourishment at national and national levels.

Figure 29: Towards attaining the MDG Indicator 4



Monitoring the WFS target

The same was observed for the monitoring of progress towards achieving the WFS target and Malawi has already achieved the target well before the year 2000 from the FAO global estimates. However, the country estimates using the 2004/5 IHS showed a different number which in between. As with the prevalence of food deprivation, it would be necessary to use the available and future IHS2 food consumption data to analyse the trends over time and at sub national levels.



IV. Conclusion

This report analyses the food security situation in Malawi using data on food consumption from the Malawi 2004/5 Integrated Household Survey, which is an important source of data for the derivation of food security indicators at national and sub national levels. The annual estimation of the MDG indicator 5 can be performed for the assessment and monitoring of the food situation at the national and sub national levels.

The National Statistical Office conducts its the IHS surveys within its integrated survey programme. The HIS is conducted after every five years, With the aim of monitoring the poverty reduction efforts within the country's growth and development strategy (MGDS) and The survey has other important information to understand the dynamics of the household economy.

It is expected that with the introduction of the FSSM, The National Statistical Office will improve on its methodology of collecting household food consumption data that can easily use the FSSM to derive FSS at the national and sub national levels and over time as well as those of others users. These FSSM are vital for the design, implementation, and evaluation and monitoring of anti hunger policies for more effective targeted interventions. The reliability and accuracy of the estimates are dependent on the accuracy of the survey in collecting food consumption data and measuring income and expenditure distribution across the population.

The limitations of the data were discussed in the executive summary, below are some of the recommendations as to increase the accuracy of the IHS which represents the best available source of data for deriving FSS estimates over time.

V: Recommendations:

The IHS2 is a comprehensive survey capturing a wide range of household data and it is essential to update the diaries for the proper collection of household food data using consumption easily identified, particularly, purchases, own consumption, obtained from relatives or as gifts, from own stocks, etc.

Food eaten away from home can be better estimated at individual level, since its mostly the one or 2 members who would eat away from home on a daily basis.

Units of measurements should be narrowed to the nearest standard unit during the household interview e.g., Basket, household should be asked what is the quantity in kgs/grammes equivalent. Alternatively the NSO should pilot that interviewers be equipped with weighing scales which they can measure all the food items that can not be easily converted such as vegetables, roots and tubers etc. A more elaborate editing procedure should be established on the field at the collection stage to avoid missing quantities or abnormal prices. Further edits should be extended at the data capture stage.

It is also recommended that Malawi develops its own detailed food composition table which will capture almost all food items that are present in Malawi which will be useful for the FSSM and for extensive food micronutrient analysis.

V. REFERENCES

1. FAO (2003). Methodology for the measurement of food deprivation. Statistics Division, Food Security Statistics. Rome. Available at the Metadata of the Food Security Statistics webpage http://www.fao.org/faostat/foodsecurity/index_en.htm
2. FAO (2006). Food Security Statistics Module, Step 1 – Processing User Manual, FAO July 2006.
3. FAO (2006). Food Security Statistics Module, Step 2 – Analysis User Manual, FAO July 2006.
4. FAO (2006). Food Security Statistics Module, Step 3 – Reports User Manual, FAO July 2006.
5. Malawi (2005). Malawi 2004/05 Integrated Household Survey. National Statistics Office. Malawi
6. Naiken L (2002). FAO Methodology for Estimating the Prevalence of Under nourishment. Key note Paper Presented at the International Scientific Symposium on Measurement of Food Deprivation and Under nutrition (26-28 June 2002).
7. Sibrián R (2006). Indicators for Monitoring Hunger at Global and Sub national levels. In: I World Congress of Public Health Nutrition Congress Proceedings. Barcelona, Spain.

8. Sibrián R Ramasawmy S and Mernies J (2007). Measuring hunger at sub national levels from household surveys using the FAO approach: MANUAL. FAO Statistics Division Working Paper Series No. ESS/ESSA/005e.

Available at the webpage. http://www.fao.org/es/ess/faostat/foodsecurity/Papers_en.htm.

9. WHO (2003). Report of a Joint WHO/FAO Expert Consultation on Diet, Nutrition and the Prevention of Chronic Diseases. WHO Technical Report Series 916. WHO, Geneva.

Available at <http://www.fao.org/FAIRDOCS/WHO/AC911E/AC911E00.HTM>.

ANNEXES:

Annex Table1. Selective Statistics of food consumption

<i>Selective statistics of food consumption</i>						
Region: Africa		Country: Malawi		Year: 2004/05		
Categories and Groupings	Number of sampled households	Average number of people in household	Average food consumption in dietary energy value (kcal/person/day)	Average food consumption in monetary value (LC\$/person/day)	Average dietary energy unit value (LC\$/1000kcal)	Average total consumption (LC\$/person/day)
Malawi	11280	4.5	2260	37.92	16.79	57.88
Income level						
Quintile 1	2256	5.9	1480	18.05	12.21	23.06
Quintile 2	2256	5.3	1710	23.84	13.92	33.05
Quintile 3	2256	4.7	2050	31.93	15.57	45.36
Quintile 4	2256	4.1	2500	43.67	17.47	63.59
Quintile 5	2256	3.2	3930	81.94	20.87	143.13
Area						
Urban	1440	4.3	2500	63.10	25.27	114.30
Rural	9840	4.6	2230	34.70	15.57	50.67
HH Size						
Up to 2 people	2195	1.6	4180	67.84	16.22	105.27
3 to 4 people	3849	3.5	2560	44.01	17.20	66.16
5 to 6 people	3177	5.4	1940	33.54	17.32	52.42
6 people and over	2059	8.2	1930	31.02	16.05	46.54
Gender of head						
Male	8697	4.7	2240	38.56	17.23	59.55
Female	2583	3.8	2350	35.21	14.99	50.83
Economic activity						
Non-agricultural activity	2253	4.5	2440	47.65	19.51	80.65
Agricultural related activity	9027	4.5	2210	35.56	16.06	52.37
Education						
No education	3050	4.3	2060	29.48	14.31	41.54
Primary	6250	4.7	2250	35.87	15.96	51.57
Secondary +	1980	4.4	2610	58.48	22.37	105.72
Region						
North	1680	4.9	2350	37.92	16.14	52.90
Centre	4320	4.7	2210	42.59	19.27	66.10
South	5280	4.3	2280	33.52	14.69	51.38
Occupation						
Farmer/mlimi	6358	4.6	2120	33.48	15.76	47.19
Employed	2916	4.4	2430	46.89	19.32	78.08
Employer/self employed	976	4.5	2400	44.81	18.68	72.10
Unemployed/others	1030	4.3	2540	35.18	13.86	57.70

Integrated Household Survey 2004/05

Annex Table 2. Food deprivation at national and sub national levels

<i>Food deprivation and parameters by population groupings</i>						
<i>Region: Africa</i>		<i>Country: Malawi</i>		<i>Year: 2004/05</i>		
Categories and Groupings	Number of sampled households	Average number of people in household	CV (%) of food dietary energy consumption (kcal/person/day) - FULL as defined by FAO	Minimum dietary energy requirement (kcal/person/day) as defined by FAO	Average of food dietary energy consumption (kcal/person/day)	Proportion of food deprivation in total population (%) as defined by FAO
Malawi	11280	4.5	41	1628	2260	27
Income level						
Quintile 1	2256	5.9	20		1480	72
Quintile 2	2256	5.3	20		1710	44
Quintile 3	2256	4.7	20		2050	14
Quintile 4	2256	4.1	20		2500	2
Quintile 5	2256	3.2	20		3930	0
Area						
Urban	1440	4.3	41	1696	2500	22
Rural	9840	4.6	42	1619	2230	28
HH Size						
Up to 2 people	2195	1.6	39	1926	4180	3
3 to 4 people	3849	3.5	33	1625	2560	11
5 to 6 people	3177	5.4	31	1580	1940	30
6 people and over	2059	8.2	47	1615	1930	43
Gender of head						
Male	8697	4.7	42	1636	2240	28
Female	2583	3.8	41	1593	2350	22
Economic activity						
Non-agricultural activity	2253	4.5	37	1646	2440	18
Agricultural related activity	9027	4.5	43	1623	2210	29
Education						
No education	3050	4.3	64	1621	2060	45
Primary	6250	4.7	36	1619	2250	22
Secondary +	1980	4.4	41	1668	2610	18
Region						
North	1680	4.9	43	1616	2350	24
Centre	4320	4.7	42	1628	2210	29
South	5280	4.3	43	1630	2280	27
Occupation						
Farmer/mlimi	6358	4.6	42	1621	2120	32
Employed	2916	4.4	37	1639	2430	18
Employer/self employed	976	4.5	44	1628	2400	24
Unemployed/others	1030	4.3	74	1633	2540	37
Integrated Household Survey 2004/05						

Annex Table 3. Share of food consumption in monetary value by food sources

<i>Share of food consumption to total consumption in monetary value and by food sources</i>						
<i>Region: Africa</i>		<i>Country: Malawi</i>		<i>Year: 2004/05</i>		
Categories and Groupings	Number of sampled households	Share of food consumption in monetary value to total consumption (%)	Share of food consumption in monetary value purchased to total food value (%)	Share of food consumption in monetary value from own production to total food value (%)	Share of food consumption in monetary value eaten away from home to total food value (%)	Share of food consumption in monetary value from other sources to total food value (%)
Nationwide	11280	65.5	45.3	43.1	1.5	10.1
Income level						
Quintile 1	2256	78.3	34.0	51.0	0.7	14.3
Quintile 2	2256	72.1	32.9	53.8	0.9	12.4
Quintile 3	2256	70.4	35.1	53.2	1.1	10.6
Quintile 4	2256	68.7	40.9	45.4	1.5	12.2
Quintile 5	2256	57.3	61.2	30.2	2.1	6.5
Area						
Urban	1440	55.2	88.3	6.3	2.5	2.9
Rural	9840	68.5	35.3	51.7	1.3	11.8
HH Size						
Up to 2 people	2195	64.4	50.4	32.6	2.7	14.3
3 to 4 people	3849	66.5	46.2	40.9	1.6	11.3
5 to 6 people	3177	64.0	45.6	43.7	1.3	9.4
6 people and over	2059	66.6	41.4	50.1	1.1	7.4
Gender of head of HH						
Male	8697	64.7	47.0	42.2	1.6	9.2
Female	2583	69.3	37.1	47.5	1.1	14.4
Age of head of HH						
14-24 years	1142	67.2	46.2	38.9	2.7	12.1
25-34 years	3444	63.4	53.6	35.8	2.1	8.5
35-49 years	3168	64.4	46.6	41.8	1.3	10.3
50-64 years	2180	67.0	37.9	53.2	1.0	7.9
65 + years	1346	72.5	28.7	53.5	0.7	17.0
Economic activity of head of HH						
Non-agricultural activity	2253	59.1	69.0	23.4	1.8	5.9
Agricultural related activity	9027	67.9	37.6	49.6	1.4	11.4
Education						
No education	3050	71.0	30.8	55.0	1.0	13.3
Primary	6250	69.6	38.7	48.2	1.5	11.6
Secondary +	1980	55.3	70.8	22.9	1.9	4.3
Region						
North	1680	71.7	37.1	53.4	1.3	8.3
Centre	4320	64.4	43.9	44.0	1.8	10.4
South	5280	65.2	49.2	39.3	1.2	10.2
Occupation						
Farmer/mlimi	6358	71.0	29.2	58.0	1.0	11.8
Employed	2916	60.1	65.1	24.8	1.8	8.2
Employer/self employed	976	62.1	67.7	25.1	3.1	4.1
Unemployed/others	1030	61.0	44.4	40.8	1.3	13.4

Malawi, Integrated Household Survey, 2004/5

Annex Table 4. Share of dietary energy consumption by food source

Table 4: Share of food dietary energy by food sources to total food dietary energy consumption

Region: Africa

Country: Malawi

Year: 2004/05

Categories and Groupings	Number of sampled households	Share of dietary energy purchased to total food consumption (%)	Share of dietary energy from own production to total food consumption (%)	Share of dietary energy eaten away from home to total food consumption (%)	Share of dietary energy from other sources to total food consumption (%)
Malawi	11280	48.5	42.0	0.8	8.7
Income level					
Quintile 1	2256	48.4	40.4	0.5	10.7
Quintile 2	2256	41.4	46.5	0.5	11.7
Quintile 3	2256	39.8	50.0	0.7	9.5
Quintile 4	2256	45.0	45.0	1.1	8.9
Quintile 5	2256	61.6	32.1	1.0	5.3
Area					
Urban	1440	88.7	7.3	1.3	2.7
Rural	9840	42.8	46.9	0.7	9.6
HH Size					
Up to 2 people	2195	53.5	31.9	1.1	13.6
3 to 4 people	3849	48.6	41.9	0.8	8.8
5 to 6 people	3177	46.6	43.4	0.7	9.3
6 people and over	2059	48.2	45.3	0.7	5.7
Gender					
Male	8697	50.1	41.1	0.9	7.9
Female	2583	42.1	45.5	0.6	11.8
Economic activity					
Non-agricultural activity	2253	68.6	24.5	0.8	6.1
Agricultural related activity	9027	43.2	46.6	0.8	9.4
Education					
No education	3050	42.3	46.0	0.5	11.2
Primary	6250	45.5	44.5	0.9	9.1
Secondary +	1980	65.4	29.3	0.9	4.4
Region					
North	1680	47.1	44.2	0.9	7.8
Centre	4320	41.9	46.5	1.1	10.5
South	5280	55.0	37.3	0.5	7.2
Occupation					
Farmer/mlimi	6358	37.0	52.6	0.7	9.7
Employed	2916	65.4	26.8	0.9	6.9
Employer/self employed	976	68.4	25.7	1.4	4.5
Unemployed/others	1030	48.9	38.9	0.6	11.6

Integrated Household Survey 2004/05

Annex Table 5. FAO indicators on food deprivation and food poverty

FAO indicators on hunger based on total consumption expenditure and the first quintile										
Region: Africa Country: Malawi Year: 2004/05										
Categories and Groupings	Prevalence of food deprivation (%)	CV of dietary energy consumption FULL as FAO (%)	Average food consumption in dietary energy value (kcal/person/day)	Minimum dietary energy requirement (kcal/person/day)	Dietary energy unit value of the first quintile (LCS/1000kcal)	Critical food poverty line of the 1st quintile	Total consumption expenditure (LCS/person/day)	CV of total consumption expenditure FULL (%)	Prevalence of Critical Food Poverty (%)	Income quintile having better balance nutritional diet
Malawi	26.7	41.5	2280	1628	13.31	21.67	57.88	78.9	14.4	3
Area										
Urban	21.9	41.4	2500	1696	13.59	23.05	114.30	102.6	7.2	1
Rural	27.9	42.3	2230	1619	13.84	22.41	50.67	63.3	13.2	4
HH Size										
Up to 2 people	2.9	38.6	4180	1926	13.31	25.65	105.27	72.9	3.3	3
3 to 4 people	10.7	33.2	2560	1625	13.31	21.63	66.16	67.3	6.4	3
5 to 6 people	30.0	30.8	1940	1590	13.31	21.03	52.42	69.0	12.5	3
6 people and over	42.7	46.7	1930	1615	13.31	21.50	46.54	71.9	19.1	3
Gender of head of HH										
Male	28.2	41.9	2240	1636	13.31	21.78	59.55	80.1	14.1	3
Female	21.6	41.1	2360	1593	13.31	21.21	50.83	73.2	15.7	3
Economic activity of head of HH										
Non-agricultural activity	18.2	37.4	2440	1646	13.31	21.91	80.65	95.1	11.1	3
Agricultural related activity	29.0	42.8	2210	1623	13.31	21.61	52.37	69.8	13.8	3
Education										
No education	45.4	64.2	2060	1621	13.31	21.58	41.54	58.6	17.5	3
Primary	22.4	36.2	2250	1619	13.31	21.56	51.57	60.7	10.1	3
Secondary +	17.6	41.5	2610	1668	13.31	22.20	105.72	92.0	5.5	3
Region										
North	23.7	42.5	2360	1616	11.27	18.20	52.90	64.2	6.4	1
Centre	29.0	42.2	2210	1628	17.10	27.84	66.10	85.1	21.1	2
South	26.8	42.6	2280	1630	14.28	23.27	51.38	76.6	20.4	2
Occupation										
Farmer/mini	32.2	42.3	2120	1621	13.31	21.59	47.19	59.6	12.6	3
Employed	17.6	36.5	2430	1639	13.31	21.83	78.08	92.3	10.9	3
Employer/self employed	23.9	44.0	2400	1628	13.31	21.67	72.10	81.9	9.4	3
Unemployed/others	36.9	74.2	2540	1633	13.31	21.75	57.70	87.6	18.0	3

Integrated Household Survey 2004/05

Annex Table 6. Contribution of in grams of nutrient consumed in 1000 calories

<i>Nutrient density per 1000 kcal</i>				
<i>Region: Africa</i>		<i>Country: Malawi</i>		<i>Year: 2004/05</i>
Food commodity group	Average food dietary energy consumption (kcal/person/day)	Protein consumption (g/1000kcal)	Carbohydrates consumption (g/1000kcal)	Fats consumption (g/1000kcal)
Malawi	2260	32.9	166.3	21.0
Income level				
Quintile 1	1480	35.6	180.4	13.9
Quintile 2	1710	33.8	175.6	16.6
Quintile 3	2050	33.3	173.3	17.7
Quintile 4	2500	32.8	165.0	21.5
Quintile 5	3930	31.0	150.7	28.8
Area				
Urban	2500	30.2	147.7	30.7
Rural	2230	33.3	169.0	19.6
HH Size				
Up to 2 people	4180	35.0	163.5	21.3
3 to 4 people	2560	32.5	166.1	21.4
5 to 6 people	1940	33.6	169.7	19.2
6 people and over	1930	31.6	164.4	22.5
Gender of head of HH				
Male	2240	33.1	165.3	21.4
Female	2350	32.0	170.6	19.6
Economic activity of head of HH				
Non-agricultural activity	2440	29.5	154.1	28.1
Agricultural related activity	2210	33.8	169.6	19.1
Education				
No education	2060	30.2	176.6	17.8
Primary	2250	34.0	166.9	20.2
Secondary +	2610	33.3	151.6	27.5
Region				
North	2350	58.6	138.2	22.4
Centre	2210	30.4	160.6	24.5
South	2280	28.6	178.7	17.6
Occupation				
Farmer/milimi	2120	34.2	172.0	17.8
Employed	2430	32.1	155.4	26.4
Employer/self employed	2400	29.7	150.7	29.6
Unemployed/others	2540	31.0	178.8	16.5
Integrated Household Survey 2004/05				

Annex Table 7. Dispersion ratios of income, food and total consumption

<i>Inequality measures (Dispersion Ratios) of food consumption, total consumption and income</i>				
<i>Region: Africa</i>		<i>Country: Malawi</i>		<i>Year: 2004/05</i>
Categories and Groupings	Dispersion ratio of food consumption in dietary energy (80/20)	Dispersion ratio of food consumption in monetary value (80/20)	Dispersion ratio of total consumption (80/20)	Dispersion ratio of income (80/20)
Nationwide	2.7	4.5	6.2	6.2
Area				
Urban	2.3	6.1	8.7	8.7
Rural	2.7	3.9	5.0	5.0
HH Size				
Up to 2 people	2.4	4.2	5.3	5.3
3 to 4 people	2.1	3.5	4.7	4.7
5 to 6 people	1.8	3.8	4.8	4.8
6 people and over	2.3	3.7	5.2	5.2
Gender of head of HH				
Male	2.7	4.6	6.3	6.3
Female	2.6	4.0	5.6	5.6
Age of head of HH				
14-24 years	2.6	4.8	5.6	5.6
25-34 years	2.6	5.0	6.0	6.0
35-49 years	2.4	4.5	6.5	6.5
50-64 years	2.5	4.1	5.9	5.9
65 + years	2.9	3.3	5.0	5.0
Economic activity of head of HH				
Non-agricultural activity	2.3	5.6	7.6	7.6
Agricultural related activity	2.7	4.1	5.5	5.5
Education				
No education	3.3	3.9	4.6	4.6
Primary	2.3	3.7	4.7	4.7
Secondary +	2.3	5.5	7.7	7.7
Region				
North	2.2	3.7	5.0	5.0
Centre	2.8	4.6	6.7	6.7
South	2.8	4.5	5.9	5.9
Occupation				
Farmer/mlimi	2.6	3.7	4.7	4.7
Employed	2.4	5.5	7.4	7.4
Employer/self employed	2.5	5.2	6.5	6.5
Unemployed/others	4.2	4.9	6.8	6.8
Malawi Integrated Household Survey 2004/5				

Annex Table 8: Inequality measures of food consumption and income. Gini coefficients

<u>Inequality measures (GINI coefficients – Free distribution) of food consumption, total consumption and income by population groupings</u>									
Region: Africa			Country: Malawi		Year: 2004/05				
Categories and Groupings	Number of sampled households	Average number of people in household	GINI of Food Dietary Energy Consumption DUE TO income (%)	GINI of Food Consumption in Monetary value) DUE TO income (%)	GINI of total consumption DUE TO income (%)	GINI of Income - FULL (%)	GINI of Food Dietary Energy Consumption - FULL (%)	GINI of Food Consumption in Monetary value) - (%)	GINI of Food Dietary Energy Consumption - FULL as defined by FAO (%)
Nationwide	11280	4.5	18.9	29.9	35.8	35.8	37.1	37.9	22.2
Area									
Urban	1440	4.3	16.7	36.9	44.5	44.5	37	38.7	22.1
Rural	9840	4.6	19.3	26.6	31.2	31.2	37.1	36.4	22.6
HH Size									
Up to 2 people	2196	1.6	15.8	28.4	34	34	38	37.5	20.8
3 to 4 people	3849	3.5	13.5	26.2	32.1	32.1	33.5	36.5	18.1
5 to 6 people	3177	5.4	11.8	27.2	32.7	32.7	30.9	34.3	16.9
6 people and over	2059	8.2	19.6	27.7	34.3	34.3	37	35.7	24.7
Gender of head of HH									
Male	8697	4.7	19.1	30.5	36.2	36.2	36.7	37.8	22.4
Female	2583	3.8	18.5	26.8	33.6	33.6	38.8	38.3	22.0
Age of head of HH									
14-24 years	1142	2.8	24.1	31.7	33.5	33.5	40.8	38.6	28.7
25-34 years	3444	4.1	18.3	31.8	35	35	35.9	37.1	21.8
35-49 years	3168	5.6	16.5	30.3	37.2	37.2	35.7	37.9	20.1
50-64 years	2180	4.9	18.3	27.6	35.5	35.5	38.5	37.7	23.1
65 + years	1346	3.8	20.6	24.5	31	31	35.8	38.2	25.8

Non-agricultural activity	2253	4.5	31.8	78.0	95.1	95.1	102.5	89.7	37.4
Agricultural related activity	9027	4.5	37.8	54.3	69.8	69.8	88.1	80.4	42.8
Education									
No education	3050	4.3	61.0	53.5	58.6	58.6	91.4	77.1	64.2
Primary	6250	4.7	30.2	49.9	60.7	60.7	87.3	77.2	36.2
Secondary +	1980	4.4	36.3	72.2	92.0	92.0	100.0	85.2	41.5
Region									
North	1680	4.9	37.5	49.2	64.2	64.2	82.5	74.3	42.5
Centre	4320	4.7	37.1	63.0	85.1	85.1	94.9	89.8	42.2
South	5280	4.3	37.7	63.5	76.6	76.6	92.2	80.8	42.6
Occupation									
Farmer/milimi	6368	4.6	37.3	49.7	59.6	59.6	79.2	75.9	42.3
Employed	2916	4.4	30.5	72.3	92.3	92.3	102.5	92.8	36.5
Employer/self employed	976	4.5	39.2	66.2	81.9	81.9	93.9	80.7	44.0
Unemployed/others	1030	4.3	71.4	63.8	87.6	87.6	125.5	83.1	74.2
Malawi Integrated Household Survey 2004/5									

Annex Table 10: Food demand elasticity with respect to income

Demand elasticity with respect to Income and population groupings

Region: Africa

Country: Malawi

Year: 2004/05

Categories and Groupings	Income deciles	Income (LC\$/person/day)	Demand elasticity of food dietary energy consumption (DEC) respect to income (%)	Demand elasticity of food consumption in monetary value (FMV) respect to income (%)	Demand elasticity of Engle ratio respect to income (%)
Income level					
Decile	1	20.65	1.1	5.7	0.9
Decile	2	24.86	0.9	2.8	0.9
Decile	3	30.58	0.7	1.8	0.9
Decile	4	35.52	0.7	1.4	0.9
Decile	5	41.96	0.6	1.1	0.9
Decile	6	48.92	0.6	1.0	0.8
Decile	7	57.65	0.5	0.8	0.8
Decile	8	70.01	0.5	0.7	0.8
Decile	9	93.12	0.4	0.6	0.8
Decile	10	199.11	0.3	0.4	0.8
Area					
Urban	1	30.04	0.8	-1297.7	0.9
Urban	2	41.43	0.6	3.1	0.9
Urban	3	43.36	0.6	2.7	0.9
Urban	4	52.10	0.5	1.8	0.9
Urban	5	60.02	0.5	1.4	0.9
Urban	6	70.85	0.5	1.2	0.9
Urban	7	94.49	0.4	0.9	0.8
Urban	8	118.09	0.4	0.7	0.8
Urban	9	174.22	0.3	0.6	0.8
Urban	10	445.34	0.2	0.4	0.8
Rural	1	19.87	1.4	3.7	0.9
Rural	2	23.96	1.1	2.2	0.9
Rural	3	29.37	0.9	1.5	0.9
Rural	4	34.20	0.8	1.2	0.9
Rural	5	40.08	0.7	1.0	0.9
Rural	6	46.86	0.6	0.9	0.9
Rural	7	54.85	0.6	0.8	0.8
Rural	8	65.30	0.5	0.7	0.8
Rural	9	83.61	0.5	0.6	0.8
Rural	10	144.65	0.4	0.4	0.8

Malawi Integrated household Survey 2004/5

Annex Table 11: Diet composition - Nutrient consumption

<i>Food consumption in monetary and nutrient values by national, sub national and population groupings</i>					
<i>Region: Africa</i>		<i>Country: Malawi</i>		<i>Year: 2004/05</i>	
Categories and Groupings	Average food dietary energy consumption (kcal/person/day)	Average food consumption in monetary value of food consumed (LC\$/person/day)	Protein consumption (g/person/day)	Carbohydrates consumption (g/person/day)	Fat consumption (g/person/day)
Malawi	2260	37.92	74.3	375.7	47.5
Income level					
Quintile 1	1480	18.05	52.5	266.5	20.6
Quintile 2	1710	23.84	57.9	300.6	28.5
Quintile 3	2050	31.93	68.3	355.3	36.4
Quintile 4	2500	43.67	81.9	412.3	53.7
Quintile 5	3930	81.94	121.7	592.0	112.9
Area					
Urban	2500	63.10	75.3	368.9	76.7
Rural	2230	34.70	74.2	376.5	43.8
HH Size					
Up to 2 people	4180	67.84	146.6	683.7	89.0
3 to 4 people	2560	44.01	83.1	424.9	54.7
5 to 6 people	1940	33.54	65.1	328.5	37.1
6 people and over	1930	31.02	61.1	317.7	43.5
Gender of head of HH					
Male	2240	38.56	74.1	369.7	47.8
Female	2350	35.21	75.3	400.9	46.2
Economic activity of head of HH					
Non-agricultural activity	2440	47.65	72.1	376.3	68.6
Agricultural related activity	2210	35.56	74.9	375.5	42.4
Education					
No education	2060	29.48	62.2	363.9	36.6
Primary	2250	35.87	76.3	375.2	45.5
Secondary +	2610	58.48	87.0	396.4	72.0
Region					
North	2350	37.92	137.7	324.6	52.6
Centre	2210	42.59	67.2	354.8	54.1
South	2280	33.52	65.4	407.9	40.1
Occupation					
Farmer/mlimi	2120	33.48	72.6	365.5	37.8
Employed	2430	46.89	77.8	377.1	64.2
Employer/self employed	2400	44.81	71.2	361.6	70.9
Unemployed/others	2540	35.18	78.7	453.8	41.9

Integrated Household Survey 2004/05

Annex Table 12: Total energy and macro nutrient by food commodity group

<i>Food consumption in quantity equivalent, monetary and nutrient values by food commodity groups</i>						
<i>Region: Africa</i>		<i>Country: Malawi</i>		<i>Year: 2004/05</i>		
Food commodity group	Average food quantity consumed (g/person/day)	Average food consumption in monetary value (LC\$/person/day)	Average food dietary energy consumption (kcal/person/day)	Average food proteins consumption (g/person/day)	Average food carbohydrates consumption (g/person/day)	Average food fats consumption (g/person/day)
CEREALS AND PRODUCTS	509.2	12.28	1104	26.3	236.5	5.2
ROOTS AND TUBERS AND PRODUCTS	155.2	1.53	188	1.5	44.2	0.4
SUGARS AND SYRUPS AND PRODUCTS	52.0	1.63	125	0.0	31.0	0.0
PULSES	52.1	2.18	175	11.2	29.5	0.7
TREE NUTS	0.0	0.00	0	0.0	0.0	0.0
OIL CROPS	39.0	1.90	152	8.5	8.6	9.0
VEGETABLES AND PRODUCTS	184.4	4.57	37	2.5	5.5	0.3
FRUITS AND PRODUCTS	73.0	1.71	37	0.6	7.1	0.6
STIMULANTS	8.5	0.86	29	2.0	4.2	0.2
SPICES	15.0	0.60	3	0.3	0.4	0.0
ALCOHOLIC BEVERAGES	46.8	1.44	38	0.9	6.4	0.0
MEAT	19.5	3.66	42	2.9	0.0	3.4
EGGS	3.1	0.61	4	0.4	0.0	0.3
FISH AND FISH PRODUCTS	60.0	2.16	85	15.9	0.0	2.4
MILK AND CHEESE	12.3	0.82	12	0.6	0.8	0.7
OILS AND FATS (vegetable oils)	23.9	1.25	215	0.0	0.0	23.8
OILS AND FATS (animal fats)	0.0	0.01	0	0.0	0.0	0.0
NON ALCOHOLIC BEVERAGES	9.0	0.34	3	0.0	0.7	0.0
MISCELLANEOUS AND PREPARED FOOD	5.1	0.37	9	0.7	0.6	0.4

Integrated Household Survey 2004/05

Annex Table 13: Share of nutrients by food commodity group

<i>Share of each nutrient in total dietary energy consumption by food commodity groups</i>				
<i>Region: Africa</i>	<i>Country: Malawi</i>		<i>Year: 2004/05</i>	
Food commodity group	Share of dietary energy consumption in Total Energy Consumption (%)	Share of protein consumption in Total Protein Consumption (%)	Share of carbohydrates Consumption in Total Carbohydrates Consumption (%)	Share of Fats Consumption in Total Fats Consumption (%)
CEREALS AND PRODUCTS	48.9	35.4	63.0	11.0
ROOTS AND TUBERS AND PRODUCTS	8.3	2.1	11.8	0.8
SUGARS AND SYRUPS AND PRODUCTS	5.5	0.0	8.3	0.0
PULSES	7.7	15.1	7.8	1.5
TREE NUTS	0.0	0.0	0.0	0.0
OIL CROPS	6.7	11.4	2.3	19.0
VEGETABLES AND PRODUCTS	1.7	3.3	1.5	0.5
FRUITS AND PRODUCTS	1.7	0.7	1.9	1.2
STIMULANTS	1.3	2.7	1.1	0.5
SPICES	0.1	0.4	0.1	0.1
ALCOHOLIC BEVERAGES	1.7	1.1	1.7	0.1
MEAT	1.9	3.9	0.0	7.1
EGGS	0.2	0.5	0.0	0.7
FISH AND FISH PRODUCTS	3.8	21.4	0.0	5.1
MILK AND CHEESE	0.5	0.8	0.2	1.5
OILS AND FATS (vegetable oils)	9.5	0.0	0.0	50.2
OILS AND FATS (animal fats)	0.0	0.0	0.0	0.1
NON ALCOHOLIC BEVERAGES	0.1	0.0	0.2	0.0
MISCELLANEOUS AND PREPARED FOOD	0.4	1.0	0.2	0.8
Integrated Household Survey 2004/05				

